Shiv Nadar University Chennai

CS3807 – Deep Learning Laboratory

Experiments 8, 9

Effect of Attention Type and Attention Position in a Pretrained ResNet50

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1. Objective

The objective of this experiment is to understand how different attention mechanisms modify a pretrained convolutional neural network and how the **position at which attention is inserted** influences image classification performance.

A single pretrained architecture,

ResNet50 pretrained on ImageNet

is used throughout the experiment.

Students will investigate:

- channel attention;
- spatial attention;
- sequential channel–spatial attention;
- lightweight self-attention;
- different positions for inserting attention within ResNet50;
- single-stage and multi-stage attention;
- attention ordering;
- additional established attention mechanisms; and
- qualitative interpretation using internal attention maps and Grad-CAM.

The experiment emphasizes that the effectiveness of attention depends not only on the attention mechanism itself, but also on the characteristics of the feature representation on which it operates.

2. Learning Outcomes

After completing this experiment, students will be able to:

1. explain the stage-wise architecture of a pretrained ResNet50;
2. distinguish early, intermediate, and late convolutional features;
3. explain the motivation behind attention mechanisms;
4. implement channel attention;
5. implement spatial attention;
6. implement sequential channel–spatial attention;
7. implement a lightweight self-attention mechanism;
8. insert an attention module at different stages of a pretrained network;
9. analyze the influence of attention position;
10. compare different attention mechanisms under a controlled experimental protocol;
11. analyze classification accuracy and class-wise performance;
12. visualize internal attention maps;
13. generate Grad-CAM explanations;
14. distinguish an internal attention map from a post-hoc explanation map;
15. analyze accuracy–complexity trade-offs; and
16. perform controlled ablation studies.

3. Dataset

Dataset: TensorFlow Flowers (`tf_flowers`)

The TensorFlow Flowers dataset contains 3,670 natural RGB images belonging to five flower classes: daisy, dandelion, roses, sunflowers, and tulips. The dataset is used to evaluate the classification performance of an ImageNet-pretrained ResNet50 backbone with different attention mechanisms.

The images exhibit variations in scale, viewpoint, illumination, background clutter, flower size, color, texture, and object location. These variations make the dataset suitable for investigating whether attention mechanisms help a model identify discriminative visual features and focus on relevant image regions.

Figure 1: Representative flower images from the TensorFlow Flowers dataset, illustrating variations in appearance, viewpoint, and background.

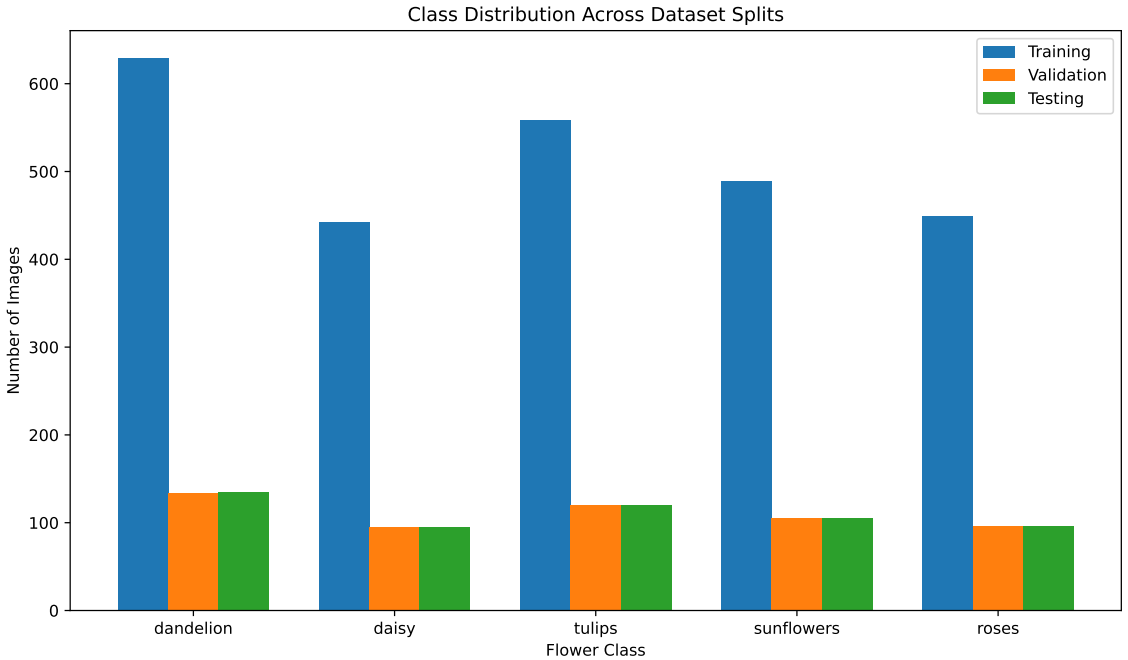

Figure 2: Class distribution across the training, validation, and test subsets. Stratified splitting helps preserve similar class proportions across the subsets.

3.1 Image Preprocessing

All input images are resized to a fixed resolution of

$$224 \times 224 \times 3,$$

where the dimensions represent image height, image width, and the three RGB channels, respectively.

The images are preprocessed using the procedure expected by the ImageNet-pretrained ResNet50 implementation. This ensures that the input representation is consistent with the preprocessing convention used when the backbone was pretrained.

The data augmentation strategy used for training includes the following transformations:

- Random horizontal flipping;
- Small random rotations;
- Random zoom;
- Random translation; and
- Mild contrast variation.

These transformations expose the model to variations in image orientation, scale, position, and appearance. They can help reduce over-reliance on particular visual patterns and improve generalization to unseen images.

Augmentation is applied during training, whereas validation and test images are used for evaluation without random training-time transformations. The same intended augmentation protocol is maintained across model comparisons so that differences in performance can be attributed more meaningfully to the model configurations.

3.2 Dataset Split

The dataset is divided into training, validation, and test subsets using a stratified split with random seed 42. The target proportions are approximately

$$\text{Training} : \text{Validation} : \text{Testing} = 70\% : 15\% : 15\%.$$

The resulting split used in the notebook is presented in Table 1.

Table 1: Dataset split used in the experiments.

Subset	**Number of Images**	**Percentage**
Training	2,569	70.00% approximately
Validation	550	15.00% approximately
Testing	551	15.00% approximately
Total	3,670	100%

The training subset is used to optimize model parameters. The validation subset is used to monitor model performance during training and support model selection, while the test subset is reserved for final evaluation.

Using the same training, validation, and test samples across the experiments reduces the influence of sample selection on measured performance differences. This is important when comparing the baseline model with different attention mechanisms and insertion positions.

4. Pretrained Architecture: ResNet50

ResNet50 pretrained on ImageNet is used as the backbone architecture throughout the experiment. The baseline model removes the original ImageNet classification head and replaces it with a task-specific classification head for the five flower classes.

For an input image

$$I \in \mathbb{R}^{224\times224\times3},$$

the backbone extracts hierarchical feature representations through successive convolutional stages. The major stages and their approximate output dimensions are shown in Table 2.

As the network becomes deeper, its spatial resolution decreases, while the number of feature channels increases:

$$\text{Spatial Resolution} \downarrow \qquad \text{Feature Channels} \uparrow$$

Early stages generally retain finer spatial information, such as edges, colors, and local textures. Intermediate stages represent more complex patterns and object parts, whereas deeper stages typically encode more abstract semantic information.

Table 2: Major ResNet50 stages and their output feature dimensions.

ResNet50 Stage	Spatial Resolution	Channels
Stem + MaxPool	56×56	64
`conv2_x`	56×56	256
`conv3_x`	28×28	512
`conv4_x`	14×14	1024
`conv5_x`	7×7	2048

These hierarchical representations make ResNet50 suitable for investigating attention mechanisms at different feature extraction stages.

4.1 Residual Learning

ResNet50 uses residual connections to facilitate the training of deep neural networks. Instead of learning an entirely new mapping, a residual block learns a residual function that is added to an identity or projected shortcut.

The residual operation can be expressed as

$$\mathbf{y} = \mathcal{F}(\mathbf{x}) + \mathbf{x},$$

where $\mathbf{x}$ represents the input feature tensor, $\mathcal{F}(\mathbf{x})$ represents the transformation learned by the residual branch, and $\mathbf{y}$ is the resulting feature representation.

When the input and output dimensions differ, a projection can be used to match their dimensions before addition.

Residual connections improve gradient propagation through deep networks and make the optimization of deep architectures more manageable. ImageNet pretraining further provides useful visual representations that can be adapted to the flower classification task.

4.2 Feature Extraction and Attention Positions

The experiment investigates attention mechanisms at different ResNet50 feature extraction stages. The following notation is used:

- P_1: output of `conv2_x`, with dimensions $56 \times 56 \times 256$;
- P_2: output of `conv3_x`, with dimensions $28 \times 28 \times 512$;
- P_3: output of `conv4_x`, with dimensions $14 \times 14 \times 1024$;
- P_4: output of `conv5_x`, with dimensions $7 \times 7 \times 2048$.

These positions provide different trade-offs between spatial detail and semantic abstraction. Attention at earlier stages operates on larger feature maps and can retain finer spatial information. Attention at deeper stages operates on more compact feature maps that generally contain richer semantic representations.

For the principal P_3 attention models, the feature tensor has dimensions

$$\mathbf{F}_{P_3} \in \mathbb{R}^{14 \times 14 \times 1024}.$$

The attention module reweights this feature representation before the remaining backbone stages and classification head are applied. The resulting features are aggregated and passed to the classifier to produce probabilities for the five flower classes.

The different attention configurations are compared against the baseline ResNet50 model to investigate their effects on classification performance, model complexity, inference time, and feature interpretability.

5. Understanding ResNet50 Feature Hierarchy

ResNet50 learns hierarchical visual representations as the input image passes through successive convolutional stages. The spatial resolution generally decreases with network depth, while the number of feature channels increases.

The approximate interpretation of the learned features is:

- **Early features:** edges, colors, orientations, boundaries, and simple textures.
- **Intermediate features:** local patterns, repeated structures, petal arrangements, flower parts, and shape information.
- **Late features:** high-level object structures and semantic representations that can help distinguish between flower classes.

These interpretations are general descriptions of hierarchical feature learning. Individual channels do not necessarily correspond to one specific visual concept.

The feature representation at each stage is different in both spatial resolution and semantic content. Consequently, the same attention mechanism may behave differently when inserted at different stages of ResNet50.

6. Why Attention Position Matters

Attention operates on the feature representation produced by the preceding network layers. Its behavior therefore depends on the dimensions and information contained in the input feature tensor.

Suppose an attention module receives

$$F \in \mathbb{R}^{H \times W \times C},$$

where H and W denote the spatial dimensions and C denotes the number of channels.

These dimensions change with network depth. Therefore, the computational cost and the type of information available to the attention module also change.

- **Early-stage attention:** operates on high-resolution features that primarily represent low-level visual patterns. Operations involving spatial relationships may be computationally expensive at this stage.
- **Middle-stage attention:** operates on intermediate representations containing spatial patterns and increasingly discriminative visual information.
- **Late-stage attention:** operates on lower-resolution features with more abstract semantic representations. Spatial operations may be less expensive because the feature maps contain fewer spatial positions.

For example, a spatial self-attention mechanism that explicitly computes relationships between all spatial positions has a computational component proportional to $(HW)^2$. Its attention matrix therefore becomes substantially larger at higher spatial resolutions.

The placement of an attention mechanism is consequently an experimental design choice rather than a universally optimal fixed location.

This experiment investigates two related questions:

Which attention mechanism should be used?

and

Where should the attention mechanism be inserted?

The answers should be determined from experimental results, including classification performance, parameter count, and computational cost.

7. Candidate Attention Positions

Four candidate attention positions are defined using the conventional ResNet50 convolutional stages.

P_1 : after conv2_x,

P_2 : after conv3_x,

P_3 : after conv4_x,

P_4 : after conv5_x.

For an input image of size 224×224, the approximate output dimensions of these stages are shown in Table 3.

For the principal attention experiments, the candidate positions P_2, P_3, and P_4 provide a useful comparison between intermediate and late feature representations.

The P_1 position may also be investigated, but its higher spatial resolution can increase the computational cost of attention mechanisms, particularly those that calculate pairwise spatial relationships.

The stage names and dimensions in this table describe the conventional ResNet50 architecture. In the implementation, each attention module must be connected to the corresponding stage output to ensure that the intended position is actually evaluated.

Table 3: Candidate attention positions in ResNet50.

Position	Location	Resolution	Channels
P_1	After conv2_x	56×56	256
P_2	After conv3_x	28×28	512
P_3	After conv4_x	14×14	1024
P_4	After conv5_x	7×7	2048

8. Baseline Model

The baseline model consists of an ImageNet-pretrained ResNet50 backbone without an additional attention mechanism, followed by a classification head for the five flower classes.

The classification head uses global average pooling, dropout, and a dense output layer.

Figure 3: Baseline ResNet50 architecture without an additional attention module.

The global average pooling layer converts the final convolutional feature maps into a one-dimensional feature vector. Dropout is used as a regularization operation before the final classification layer.

For a five-class classification task, the output layer contains five units. When a softmax activation is used, the output can be interpreted as a probability distribution over the five flower classes.

The baseline establishes the reference performance against which attention-based models are evaluated. All comparisons should use consistent data splits and evaluation procedures.

9. Channel Attention

Channel attention estimates the relative importance of different feature channels. It recalibrates the feature representation by assigning a learned weight to each channel.

Consider an input feature tensor

$$F \in \mathbb{R}^{H \times W \times C}.$$

First, global average pooling is applied across the spatial dimensions to generate a channel descriptor:

$$z_c = \frac{1}{HW} \sum_{i=1}^{H} \sum_{j=1}^{W} F(i, j, c), \qquad c = 1, \ldots, C.$$

The resulting descriptor is

$$z \in \mathbb{R}^C.$$

The descriptor is passed through a bottleneck transformation consisting of two learnable fully connected transformations, with a nonlinear activation between them. The channel-attention weights are calculated as

$$M_c(F) = \sigma\left(W_2 \delta(W_1 z)\right),$$

where:

- $\delta(\cdot)$ denotes a nonlinear activation, such as ReLU;
- $\sigma(\cdot)$ denotes the sigmoid activation;
- W_1 reduces the descriptor dimension from C to C/r; and
- W_2 restores the representation to C dimensions.

The reduction ratio is defined as

$$r = 16.$$

Thus, the intermediate bottleneck dimension is $C/16$, subject to the implementation's handling of integer dimensions.

The channel-attention vector has dimension

$$M_c(F) \in \mathbb{R}^C.$$

It is broadcast across the spatial dimensions and multiplied element-wise with the original feature tensor:

$$F_c = F \odot M_c(F).$$

Here, $\odot$ denotes element-wise multiplication with broadcasting.
The refined tensor F_c retains the same dimensions as F, but its channels are rescaled according to their learned attention weights.
Channel attention therefore attempts to answer:

Which feature channels are important?

In the flower-classification task, channel attention may help emphasize feature channels useful for distinguishing visual characteristics such as petal patterns, textures, and flower structures. Whether this improves classification performance must be verified experimentally.
The reduction ratio $r = 16$ is used as the default setting; it is not treated as a separate mandatory ablation in this experiment.

10. Spatial Attention

Spatial attention estimates the relative importance of different locations in a feature map. Unlike channel attention, which assigns weights to channels, spatial attention generates a spatial attention map that is shared across channels.
Consider the input feature tensor

$$F \in \mathbb{R}^{H \times W \times C}.$$

First, average pooling and maximum pooling are performed across the channel dimension.
The average-pooled feature map is

$$F_{\text{avg}}(i, j) = \frac{1}{C} \sum_{c=1}^{C} F(i, j, c),$$

and the maximum-pooled feature map is

$$F_{\max}(i, j) = \max_{1 \le c \le C} F(i, j, c).$$

Both pooled feature maps have dimensions

$$F_{\text{avg}}, F_{\max} \in \mathbb{R}^{H \times W \times 1}.$$

The two maps are concatenated along the channel dimension:

$$F_{\text{pool}} = [F_{\text{avg}} \| F_{\max}],$$

where $\|$ denotes concatenation. Consequently,

$$F_{\text{pool}} \in \mathbb{R}^{H \times W \times 2}.$$

A convolution with a 7×7 kernel is then applied to the concatenated feature map, followed by a sigmoid activation:

$$M_s(F) = \sigma\left(\text{Conv}_{7\times7}(F_{\text{pool}})\right).$$

With appropriate padding, the resulting spatial attention map has dimensions

$$M_s(F) \in \mathbb{R}^{H \times W \times 1}.$$

The spatial attention map is broadcast across the channel dimension and multiplied element-wise with the original feature tensor:

$$F_s = F \odot M_s(F).$$

The refined tensor retains the original dimensions:

$$F_s \in \mathbb{R}^{H \times W \times C}.$$

Spatial attention therefore attempts to answer:

$$\boxed{\text{Which spatial locations are important?}}$$

For flower classification, spatial attention may emphasize image regions containing discriminative visual information, such as petals or flower centers. However, the attention map alone does not prove that the model is focusing on semantically meaningful regions.

The 7×7 convolution is the default spatial-attention setting described here. Other kernel sizes can be compared experimentally in the dedicated kernel-size ablation, provided the corresponding models are implemented and evaluated.

11. Sequential Channel–Spatial Attention

Channel and spatial attention can be applied sequentially to refine a feature representation. In this experiment, channel attention is applied first, followed by spatial attention.

Let the input feature tensor be

$$F \in \mathbb{R}^{H \times W \times C}.$$

First, channel attention generates channel-wise importance weights and recalibrates the input features:

$$F_c = M_c(F) \odot F,$$

where

$$M_c(F) \in \mathbb{R}^{1 \times 1 \times C}.$$

The channel weights are broadcast across the spatial dimensions before element-wise multiplication.

Next, spatial attention operates on the channel-refined feature tensor:

$$F_{cs} = M_s(F_c) \odot F_c,$$

where

$$M_s(F_c) \in \mathbb{R}^{H \times W \times 1}.$$

The spatial weights are broadcast across the channel dimension. Therefore, the final feature tensor retains the dimensions of the original input:

$$F_{cs} \in \mathbb{R}^{H \times W \times C}.$$

The overall structure is

$$\boxed{F \rightarrow \text{Channel Attention} \rightarrow \text{Spatial Attention} \rightarrow F_{cs}}$$

Channel attention emphasizes the relative importance of feature channels, while spatial attention emphasizes the relative importance of spatial locations.

This sequential arrangement can be interpreted as first asking

"What is important?"

and then asking

"Where is it important?"

These descriptions represent the intended roles of the two modules; they do not guarantee that the learned weights correspond to human-interpretable concepts.

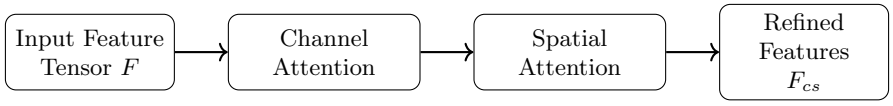

Figure 4: Sequential channel–spatial attention. Spatial attention operates on the output of channel attention.

Figure 4 illustrates the sequence.

The channel–spatial attention block can be inserted at a selected ResNet50 stage, such as P_3. In that case, the feature tensor produced by conv4_x is refined before being passed to conv5_x.

The effect of this combination should be evaluated against the baseline and the individual channel-attention and spatial-attention models under the same evaluation conditions.

12. Self-Attention

Self-attention enables spatial positions in a feature map to exchange information directly. Unlike channel attention and spatial attention, which generate channel-wise or spatial-wise importance weights, self-attention calculates relationships between pairs of spatial positions.

Consider a feature tensor

$$F \in \mathbb{R}^{H\times W\times C}.$$

First, flatten the spatial dimensions to obtain

$$X \in \mathbb{R}^{N\times C},$$

where

$$N = HW.$$

Learnable transformations generate the query, key, and value representations:

$$Q = XW_Q,$$

$$K = XW_K,$$

$$V = XW_V.$$

Let d denote the query and key dimension. The spatial attention matrix is calculated using scaled dot-product attention:

$$A = \text{softmax}\left(\frac{QK^T}{\sqrt{d}}\right).$$

The softmax is applied across the key positions for each query position. Consequently,

$$A \in \mathbb{R}^{N\times N}.$$

Each element of A represents the normalized attention weight between a query position and a key position.

The attended representation is

$$Y = AV.$$

If $V \in \mathbb{R}^{N\times d_v}$, then

$$Y \in \mathbb{R}^{N\times d_v}.$$

To combine the attended representation with the original feature tensor through a residual connection, the output must be transformed to a compatible dimension. Let W_O denote an optional output projection when required:

$$Y_O = YW_O.$$

For a residual connection with the original representation,

$$Y_O \in \mathbb{R}^{N \times C}.$$

After reshaping the output into the original spatial structure, a residual formulation can be expressed as

$$F_{\text{SA}} = F + \gamma \,\text{reshape}(Y_O),$$

where γ is a learnable scaling parameter.

The output projection is only required when the attended representation does not already have C channels. The exact transformations must match the implementation used in the experiment.

The residual connection preserves a direct path from the original features to the output. When γ is initialized to zero, the module initially produces the original feature tensor, assuming the residual branch has no additional effect outside this formulation.

Self-attention therefore attempts to answer:

Which spatial features are related to other spatial features?

For example, information from one spatial location can contribute to the representation at another location, even when those locations are far apart in the feature map. This provides a mechanism for capturing long-range spatial relationships.

Computational Considerations

The attention matrix contains N^2 entries, where

$$N = HW.$$

Therefore, its memory requirement is proportional to

$$O(N^2) = O\left((HW)^2\right).$$

The query–key matrix multiplication also has a computational cost proportional to $N^2 d$, in addition to the costs of computing the query, key, and value projections.

For the candidate ResNet50 positions, the number of spatial positions is:

$$\begin{aligned} P_1 &: 56 \times 56 = 3136, \\ P_2 &: 28 \times 28 = 784, \\ P_3 &: 14 \times 14 = 196, \\ P_4 &: 7 \times 7 = 49. \end{aligned}$$

The corresponding attention-matrix sizes are shown in Table 4.

Table 4: Spatial attention-matrix sizes at candidate ResNet50 stages.

Position	**Resolution**	**Spatial Positions N**	**Entries in $N \times N$**
P_1	56×56	3136	9,834,496
P_2	28×28	784	614,656
P_3	14×14	196	38,416
P_4	7×7	49	2,401

The table shows that the spatial attention matrix becomes substantially smaller at deeper stages. In particular, the matrix at P_1 contains far more entries than the matrix at P_3 or P_4.

Therefore, P_3 and P_4 are practical candidate positions for self-attention in this experiment. However, the final choice should also consider classification performance, memory usage, parameter count, and inference time.

13. Inserting Attention into ResNet50

An attention module can be inserted after a selected ResNet50 stage to refine the features before they are passed to subsequent layers.

For a selected stage P_k, the general structure is

$$\text{Previous ResNet Stage} \rightarrow \text{Attention} \rightarrow \text{Next ResNet Stage.}$$

For example, inserting an attention module after conv4_x gives

$$\boxed{\text{conv4_x} \rightarrow \text{Attention} \rightarrow \text{conv5_x}}$$

For sequential channel–spatial attention, the structure becomes

$$\boxed{\begin{aligned}\text{conv4_x} &\rightarrow \text{Channel Attention}\\ &\rightarrow \text{Spatial Attention}\\ &\rightarrow \text{conv5_x.}\end{aligned}}$$

The same principle applies to the other candidate positions, provided the attention block is connected to the intended stage output.

Feature-Dimension Consistency

Suppose the selected stage produces

$$F_k \in \mathbb{R}^{H_k \times W_k \times C_k}.$$

For an attention module intended only to refine the features, the expected output is

$$\widetilde{F}_k \in \mathbb{R}^{H_k \times W_k \times C_k}.$$

Thus,

$$\boxed{\text{shape}(\widetilde{F}_k) = \text{shape}(F_k)}$$

for channel attention, spatial attention, sequential channel–spatial attention, and a residual self-attention block that preserves the original feature dimensions.

An implementation may use internal projections with different dimensions, but the final output must be compatible with the next ResNet50 stage. If a mechanism intentionally changes the feature dimensions, an appropriate projection or architectural adjustment is required.

Implementation Checks

For each attention-based model, the following should be verified:

- The attention module is connected to the intended ResNet50 stage output.
- The feature dimensions immediately before and after the attention block are recorded.
- The output is compatible with the subsequent ResNet50 stage and the classification head.
- The baseline and attention-based models use consistent training, validation, and test splits.
- Parameter counts, classification metrics, and inference time are recorded using the same measurement procedure.

These checks help ensure that performance differences can be interpreted in relation to the attention mechanism rather than unintended changes in the model architecture or evaluation procedure.

14. Main Experiment 1: Attention-Type Ablation

The first experiment investigates how different attention mechanisms affect flower-classification performance. To make the comparison meaningful, the attention position is fixed at

$$\boxed{P_3 = \text{after conv4_x}}$$

for all attention-based models.

The baseline model contains no additional attention module. The attention-based models are defined as shown in Table 5.

For A3, channel attention is applied first, followed by spatial attention:

$$F \rightarrow M_c \rightarrow M_s.$$

Table 5: Attention-type ablation at position P_3.

Model	Channel	Spatial	Self-Attention
A0: Baseline ResNet50	–	–	–
A1: Channel Attention	✓	–	–
A2: Spatial Attention	–	✓	–
A3: Channel–Spatial Attention	✓	✓	–
A4: Self-Attention	–	–	✓
A5: Combined Attention	✓	✓	✓

For A4, the feature tensor is refined using self-attention.

For A5, the combination of channel, spatial, and self-attention must follow the order implemented in the model. If the three modules are applied sequentially, the architecture should be documented explicitly. For example, one possible arrangement is

$$F \rightarrow M_c \rightarrow M_s \rightarrow \text{Self-Attention}.$$

This ordering is a design specification, not a claim that it was necessarily used in the executed notebook. The A5 label should be retained only if the corresponding combined model is implemented and evaluated.

The principal research question is

How does attention type affect classification performance?

For a controlled comparison, the following experimental settings should be held constant wherever applicable:

- Training, validation, and test data splits.
- Input size and preprocessing.
- Data augmentation.
- Backbone architecture and pretrained weights.
- Classification head and dropout configuration.
- Optimizer and learning-rate schedule.
- Batch size and training protocol.
- Evaluation metrics and inference-time measurement procedure.

The attention module is the intended variable of interest. Any necessary differences in parameter count or computational cost should be reported rather than assumed to be identical.

The results should be compared using test accuracy, macro F1-score, parameter count, and inference time. These measurements help assess both predictive performance and computational efficiency.

15. Main Experiment 2: Attention-Position Ablation

The second experiment investigates the influence of attention placement within ResNet50.

The attention type is fixed as sequential channel–spatial attention:

Channel Attention → Spatial Attention

The same attention design is inserted after different ResNet50 stages. The candidate positions and feature dimensions are listed in Table 6.

Table 6: Attention-position ablation using channel–spatial attention.

Experiment	Attention Position	Resolution	Channels
B1	After conv2_x (P_1)	56×56	256
B2	After conv3_x (P_2)	28×28	512
B3	After conv4_x (P_3)	14×14	1024
B4	After conv5_x (P_4)	7×7	2048

The purpose is to investigate whether attention is more effective when applied to:

- High-resolution features containing relatively low-level visual information.
- Intermediate representations that combine spatial patterns with increasingly discriminative features.
- Low-resolution representations containing higher-level semantic information.

The principal research question is

How does the location of attention influence performance?

The attention mechanism should have the same mathematical design at every position. However, its input dimensions necessarily differ between stages. Therefore, the resulting parameter count and computational cost may also differ.
The comparisons should use the same data splits, preprocessing, evaluation metrics, and training protocol. Any differences in fine-tuning configuration between stage-specific models should be documented because they can affect the comparison.

16. Main Experiment 3: Multiple Attention Positions

Attention can be applied at more than one stage of ResNet50. This experiment investigates whether refining multiple feature representations provides an advantage over applying attention at a single stage.
The primary configurations are:

$$\text{Attention only at } P_3,$$

$$\text{Attention only at } P_4,$$

and

$$\text{Attention at } P_3 + P_4.$$

An extended configuration may also be considered:

$$P_2 + P_3 + P_4.$$

For a fair comparison, the same attention mechanism should be used in each configuration. The locations at which the modules are inserted must be recorded explicitly.
Table 7 provides a template for documenting the experiment.

Table 7: Experimental configurations for multiple attention positions.

Configuration	P_2	P_3	P_4
C1: P_3 only	–	✓	–
C2: P_4 only	–	–	✓
C3: $P_3 + P_4$	–	✓	✓
C4: $P_2 + P_3 + P_4$ (extended)	✓	✓	✓

Applying attention at additional stages can increase the number of parameters and the computational work performed during training and inference. It may also increase memory requirements.
The additional modules should therefore be evaluated in terms of:

- Classification accuracy and macro F1-score.
- Total and trainable parameter counts.
- Inference time.
- Peak memory usage, if measured.
- Training time, if recorded consistently.

The main research question is

Does attention at more locations justify its additional cost?

A model with more attention modules is not necessarily more accurate. The outcome must be determined from the experimental results.

17. Main Experiment 4: Attention Order

When channel and spatial attention are combined, the order of application may affect the refined feature representation.

This experiment compares the following arrangements at the same insertion position:

$$\boxed{\text{Channel} \rightarrow \text{Spatial}}$$

and

$$\boxed{\text{Spatial} \rightarrow \text{Channel}}$$

Let the original feature tensor be

$$F \in \mathbb{R}^{H \times W \times C}.$$

For channel-first attention, the intermediate and final features are calculated as

$$F_c = M_c(F) \odot F,$$

$$F_{cs} = M_s(F_c) \odot F_c.$$

For spatial-first attention, the operations are reversed:

$$F_s = M_s(F) \odot F,$$

$$F_{sc} = M_c(F_s) \odot F_s.$$

In both cases, the output is expected to retain the dimensions of the input tensor:

$$F_{cs}, F_{sc} \in \mathbb{R}^{H \times W \times C}.$$

The two arrangements use the same types of attention modules, but the second module receives a different intermediate feature representation. Consequently, their outputs need not be identical.

The principal question is whether selecting informative channels before refining spatial locations produces different results from refining spatial locations first.

For a controlled comparison, the following should remain consistent:

- Attention insertion position.
- Attention module definitions and hyperparameters.
- Dataset splits and preprocessing.
- Training protocol and classification head.
- Evaluation metrics and inference-time procedure.

The experiment should compare test accuracy, macro F1-score, parameter count, and inference time. Any observed performance difference should be interpreted in relation to these measurements.

18. Main Experiment 5: Spatial Attention Kernel Size

The spatial-attention kernel size controls the local neighborhood used to generate the spatial attention map. Different kernel sizes may capture different amounts of spatial context.

The experiment evaluates the following kernel sizes:

$$k \in \{3, 5, 7\}.$$

The attention position, channel-pooling operations, training protocol, and evaluation procedure should remain consistent across the configurations.

Table **??** presents the recorded classification accuracy for each kernel size.

Table 8: Spatial-attention kernel-size ablation on the test set.

Spatial Kernel	**Accuracy**	**Macro F1**	**Inference Time (ms/image)**
3×3	0.9274	0.9263	7.6773
5×5	0.9310	0.9304	7.7349
7×7	0.9310	0.9304	7.7956

The accuracy increases from 92.74% with the 3×3 kernel to 93.10% with the 5×5 kernel, an improvement of 0.36 percentage points. Increasing the kernel size from 5×5 to 7×7 does not provide a further measured accuracy improvement.

The recorded results therefore suggest that the 5×5 kernel offers a useful balance between classification accuracy and inference latency in this experiment. Although the 7×7 kernel achieves the same accuracy, its measured latency is slightly higher.

The macro F1-score, parameter counts, and exact inference times should be added only if the corresponding numerical measurements are available.

The principal research question is:

Does a larger spatial kernel provide a useful performance gain?

For this run, increasing the kernel from 3×3 to 5×5 improves accuracy, whereas increasing it further to 7×7 does not improve accuracy. The conclusion is limited to the evaluated models and training conditions.

19. Additional Task: Comparison of Different Attention Mechanisms

This experiment compares the performance and computational cost of different attention mechanisms using an ImageNet-pretrained ResNet50 backbone for five-class flower classification.

The additional attention mechanisms evaluated are:

- Squeeze-and-Excitation (SE) Attention;
- Efficient Channel Attention (ECA); and
- Convolutional Block Attention Module (CBAM).

These mechanisms are compared with the baseline model and the channel, spatial, channel–spatial, and self-attention models.

For a controlled comparison, the backbone remains ResNet50. The additional attention mechanisms are evaluated at P_3, after the conv4_x stage, as specified in the experimental setup.

19.1 Comparison of Attention Mechanisms

Table 9 summarizes the recorded results from the executed notebook. Accuracy, precision, recall, and macro F1 are reported on a scale from 0 to 1. Parameter counts include the complete classification model, and inference time is reported in milliseconds per image.

Table 9: Comparison of attention mechanisms using the recorded Section 19 notebook results.

Attention Mechanism	Accuracy	Precision	Recall	Macro F1	Total Parameters	Inference Time (ms/image)
No Attention	0.9347	0.9350	0.9332	0.9339	23,597,957	7.5523
Channel Attention	0.9401	0.9409	0.9376	0.9389	23,730,117	7.8073
Spatial Attention	0.9292	0.9275	0.9293	0.9283	23,598,056	7.6747
Channel + Spatial	0.9201	0.9196	0.9206	0.9194	23,730,216	7.8358
Self-Attention	0.9347	0.9360	0.9331	0.9343	24,909,958	7.9642
SE	0.9310	0.9300	0.9312	0.9302	23,730,117	7.8236
ECA	0.9201	0.9191	0.9226	0.9201	23,597,960	7.8548
CBAM	0.9238	0.9246	0.9220	0.9228	23,730,216	8.1261

Note: Precision, recall, and macro F1 are macro-averaged across the five flower classes. Inference times are measurements from the notebook environment and should not be interpreted as hardware-independent benchmarks.

19.2 Analysis of the Results

The results show that channel attention achieved the highest recorded accuracy of 0.9401 and macro F1-score of 0.9389. Compared with the baseline accuracy of 0.9347, this corresponds to an improvement of 0.54 percentage points.

Spatial attention achieved an accuracy of 0.9292, while the channel–spatial model achieved 0.9201. Thus, combining channel and spatial attention did not improve classification performance in this particular run.

Self-attention achieved the same accuracy as the baseline, 0.9347, with a higher parameter count of $24,909,958$ compared with $23,597,957$ for the baseline. This suggests that the additional complexity of self-attention did not produce a measurable accuracy improvement in this experiment.

Among the additional mechanisms, SE achieved an accuracy of 0.9310, ECA achieved 0.9201, and CBAM achieved 0.9238. CBAM recorded precision of 0.9246, recall of 0.9220, and macro F1 of 0.9228. It also had the highest recorded inference time among the compared mechanisms, at 8.1261 ms/image.

These observations apply to the recorded experimental run. They do not establish that one attention mechanism is universally better, because results may depend on training conditions, initialization, and implementation details.

19.3 Answers to the Comparison Questions

1. **How does classification accuracy vary across attention mechanisms?**
   Channel attention achieved the highest recorded accuracy of 0.9401. The baseline and self-attention models both achieved 0.9347, while the other attention models had lower accuracy in this run.
2. **How does macro F1-score vary across attention mechanisms?**
   Channel attention achieved the highest recorded macro F1-score of 0.9389. The baseline achieved 0.9339, while the channel–spatial model achieved 0.9194.
3. **How many parameters are introduced by each attention module?**
   The total parameter count ranged from $23,597,957$ for the baseline to $24,909,958$ for self-attention. The self-attention model therefore had $1,312,001$ more total parameters than the baseline.
4. **How does each attention mechanism influence inference time?**
   The baseline had a recorded inference time of 7.5523 ms/image. CBAM had the highest recorded time, at 8.1261 ms/image. The measured times are specific to the notebook's execution environment.
5. **Does attention improve convergence behavior?**
   The final evaluation table alone is insufficient to determine whether attention improves convergence. Training and validation loss curves are required to compare convergence behavior.
6. **Does attention produce more focused feature localization?**
   Classification metrics alone cannot establish whether feature localization is more focused. Attention maps and Grad-CAM visualizations should be inspected for the same test images.
7. **Does greater attention complexity always improve classification performance?**
   No. In this experiment, self-attention introduced more parameters without improving accuracy over the baseline. The channel–spatial model also performed below the baseline.
8. **Can lightweight attention provide competitive performance with a more computationally expensive mechanism?**
   The results depend on the mechanism. ECA had a parameter count close to that of the baseline, but its recorded accuracy was lower than that of channel attention. Therefore, parameter efficiency alone does not guarantee better classification performance.

19.4 Overall Observation

The experiment demonstrates that attention mechanisms should be evaluated using predictive performance and computational cost together. Channel attention provided the strongest recorded accuracy and macro F1-score in this run. However, the results also show that adding more complex attention operations does not necessarily improve performance.

A complete assessment should consider classification metrics, parameter count, inference time, training curves, and visual evidence from attention maps and Grad-CAM. Repeated runs would also be needed to establish whether the observed differences are consistent.

20. Transfer-Learning Strategy

The experiment uses ResNet50 pretrained on ImageNet as the feature-extraction backbone. The original ImageNet classification head is removed by setting `include_top=False`. A five-class classification head is then added for the TensorFlow Flowers dataset.

The model consists of the following components:

1. Load ResNet50 using ImageNet pretrained weights.
2. Remove the original ImageNet classification head.
3. Insert the selected attention module at the required feature stage, when applicable.
4. Apply Global Average Pooling.
5. Apply Dropout with a rate of 0.3.
6. Add a Dense output layer with five neurons and softmax activation.
7. Initially freeze the pretrained feature-extraction layers and train the newly introduced layers.
8. Fine-tune the selected final convolutional layers using a smaller learning rate.

The initial training stage uses the Adam optimizer with a learning rate of

$$\eta_{\text{initial}} = 10^{-3}.$$

During fine-tuning, the learning rate is reduced to

$$\eta_{\text{fine}} = 10^{-5}.$$

In the implementation, the final convolutional stage is selected for fine-tuning by setting the corresponding `conv5_` layers as trainable. Batch-normalization layers are kept non-trainable during this stage.

The notebook uses early stopping and model checkpointing in the attention experiments where specified. The training protocol and number of trainable layers should be considered when interpreting comparisons between experiments, particularly when attention is inserted at different ResNet50 stages.

21. Loss Function

The five-class flower classification problem is trained using sparse categorical cross-entropy. This loss is appropriate because the target labels are represented as integer class indices rather than one-hot encoded vectors.

For a dataset containing N samples and $C = 5$ classes, the loss is

$$\mathcal{L}_{\text{SCE}} = -\frac{1}{N}\sum_{i=1}^{N} \log\left(\widehat{p}_{i,y_i}\right),$$

where y_i is the integer ground-truth class label for sample i, and $\widehat{p}_{i,y_i}$ is the predicted probability assigned to that correct class.

The model produces five class probabilities using the softmax function:

$$\widehat{p}_{i,c} = \frac{\exp(z_{i,c})}{\sum_{j=1}^{5} \exp(z_{i,j})},$$

where $z_{i,c}$ is the logit for class c.

The same loss formulation is used in the baseline and attention model implementations. Sparse categorical cross-entropy is equivalent to categorical cross-entropy when the latter is computed using the corresponding one-hot encoded labels.

22. Evaluation Metrics

The models are evaluated on the held-out test set using accuracy, macro precision, macro recall, macro F1-score, parameter count, and average inference time per image.

22.1 Classification Accuracy

Accuracy measures the proportion of correctly classified test images:

$$\text{Accuracy} = \frac{\text{Number of Correct Predictions}}{\text{Total Number of Test Samples}}.$$

22.2 Precision and Recall

For class c, precision and recall are defined as

$$\text{Precision}_c = \frac{TP_c}{TP_c + FP_c},$$

$$\text{Recall}_c = \frac{TP_c}{TP_c + FN_c},$$

where TP_c, FP_c, and FN_c denote true positives, false positives, and false negatives for class c.

22.3 F1-Score

The F1-score for class c is the harmonic mean of its precision and recall:

$$F1_c = 2\frac{\text{Precision}_c \text{Recall}_c}{\text{Precision}_c + \text{Recall}_c}.$$

Macro F1 is calculated by averaging the class-wise F1-scores:

$$\text{Macro F1} = \frac{1}{C}\sum_{c=1}^{C} F1_c, \qquad C = 5.$$

Macro precision and macro recall are calculated in the same way, by averaging the corresponding class-wise metrics. In the notebook, these metrics are computed using the scikit-learn functions with `average="macro"` and `zero_division=0`.

22.4 Parameter Count

The total number of parameters is obtained using the Keras `count_params()` method. The trainable parameter count is computed from the model's trainable variables.

These measurements help compare the model complexity introduced by different attention mechanisms.

22.5 Inference Time

Inference time is measured by running the model on the test batches with training disabled. The elapsed time is divided by the total number of test images and converted to milliseconds:

$$T_{\text{image}} = \frac{T_{\text{elapsed}}}{N_{\text{test}}} \times 1000 \quad \text{ms/image}.$$

Inference time depends on the hardware, runtime environment, batch processing, and model implementation. The reported values should therefore be interpreted as measurements from the notebook environment rather than universal model runtimes.

23. Required Main Result Table

Table 10 summarizes the recorded evaluation results for the baseline and attention-based models. Precision, recall, and F1 are macro-averaged. Inference time is reported in milliseconds per image.

Table 10: Main test-set comparison of ResNet50 attention models.

Model	Position	Accuracy	Precision	Recall	Macro F1	Total Parameters	Inference Time (ms/image)
Baseline	–	0.9347	0.9350	0.9332	0.9339	23,597,957	7.5673
Channel Attention	P_3	0.9401	0.9409	0.9376	0.9389	23,730,117	7.6675
Spatial Attention	P_3	0.9292	0.9275	0.9293	0.9283	23,598,056	7.6061
Channel + Spatial	P_2	0.9111	0.9090	0.9151	0.9106	23,631,368	7.8450
Channel + Spatial	P_3	0.9201	0.9196	0.9206	0.9194	23,730,216	7.6711
Channel + Spatial	P_4	0.9220	0.9196	0.9222	0.9201	24,124,520	7.7778
Self-Attention	P_3	0.9347	0.9360	0.9331	0.9343	24,909,958	7.8668
Squeeze-and-Excitation (SE)	P_3	0.9310	0.9300	0.9312	0.9302	23,730,117	7.5926
Efficient Channel Attention (ECA)	P_3	0.9201	0.9191	0.9226	0.9201	23,597,960	7.8403

23.1 Interpretation of Results

Channel attention at P_3 achieved the highest accuracy (0.9401) and macro F1-score (0.9389) among the nine models listed in Table 10. Its accuracy was approximately 0.54 percentage points higher than the baseline accuracy of 0.9347.

Spatial attention at P_3 achieved an accuracy of 0.9292, while channel-plus-spatial attention at P_3 achieved 0.9201. The channel-plus-spatial models at P_2, P_3, and P_4 achieved accuracies of 0.9111, 0.9201, and 0.9220, respectively. These results indicate that the insertion position can influence classification performance, although differences in the trainable layers and fine-tuning procedure should also be considered.

The self-attention model achieved the same recorded accuracy as the baseline (0.9347), but used more parameters and had a longer measured inference time. Thus, increasing model complexity did not automatically improve classification accuracy in this run.

Among the additional channel-attention mechanisms, SE achieved an accuracy of 0.9310, while ECA achieved 0.9201. The results suggest that the effect of an attention mechanism depends on its design and cannot be determined from its complexity alone.

These values are single-run measurements. Repeated runs with different random seeds would be required to assess the variability and statistical significance of the differences.

23.2 Additional Attention Mechanism: CBAM

The notebook also evaluates the Convolutional Block Attention Module (CBAM). Its recorded results are summarized below.

Table 11: CBAM test-set results.

Metric	**Recorded Value**
Accuracy	0.9238
Precision (macro)	0.9246
Recall (macro)	0.9220
Macro F1-score	0.9228
Total parameters	23,730,216
Inference time (ms/image)	8.1261

CBAM combines channel and spatial attention sequentially. In this run, it achieved an accuracy of 0.9238, which was lower than the baseline and the channel-attention model. It also had the longest measured inference time among the additional mechanisms listed in the notebook comparison.

The comparison demonstrates that attention modules should be evaluated using classification performance and computational cost together, rather than assuming that a more complex attention design will necessarily produce better results.

“‘latex

24. Graphical Performance Analysis

The performance of the baseline ResNet50 and attention-based models was analyzed using training curves, classification metrics, model complexity, and inference time. The graphs were generated using Matplotlib and saved in the `Figures/` directory.

24.1 Training and Validation Accuracy

Figure 5 compares training and validation accuracy across epochs for the baseline and attention-based models. Solid lines represent training accuracy, while dashed lines represent validation accuracy.

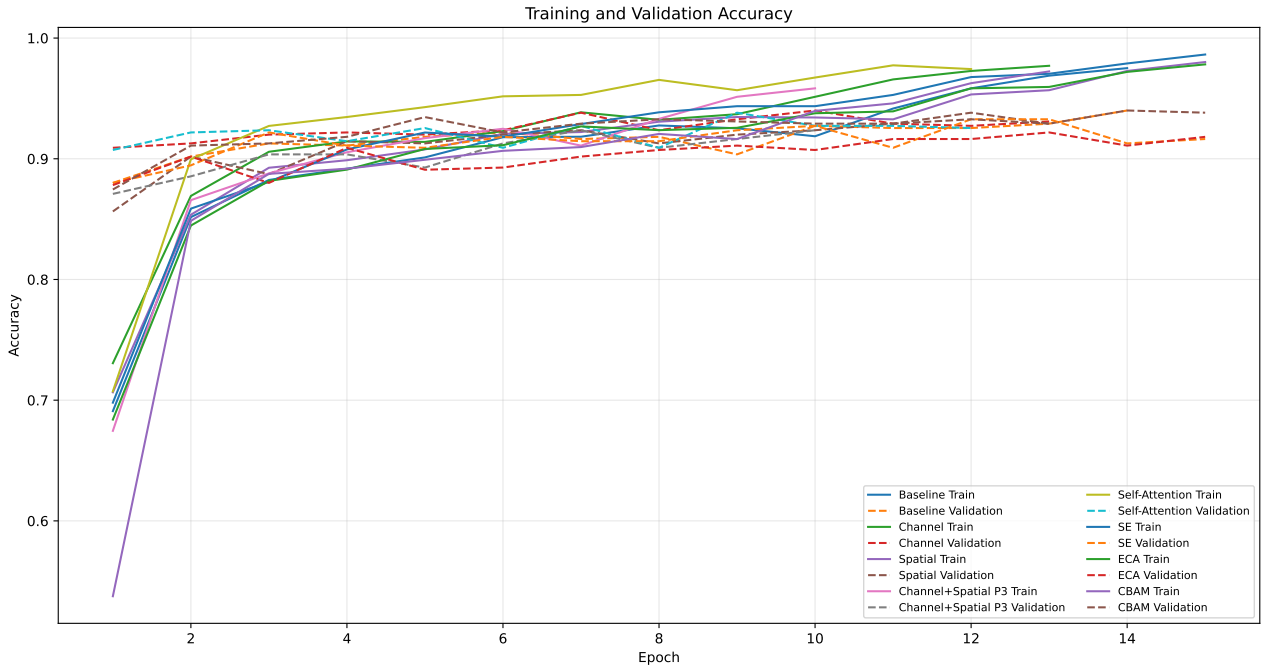

Figure 5: Training and validation accuracy of the baseline and attention-based ResNet50 models.

The training and validation curves provide information about convergence and generalization. A model whose training accuracy increases while validation accuracy stops improving may be overfitting. A large gap between the two curves can also indicate limited generalization.

The plotted curves should be examined to compare the convergence behavior of the different attention mechanisms. The final test accuracy alone is not sufficient to establish that an attention mechanism accelerates convergence or produces smoother training.

24.2 Training and Validation Loss

Figure 6 presents the training and validation categorical cross-entropy loss across epochs for the models included in the experiment.

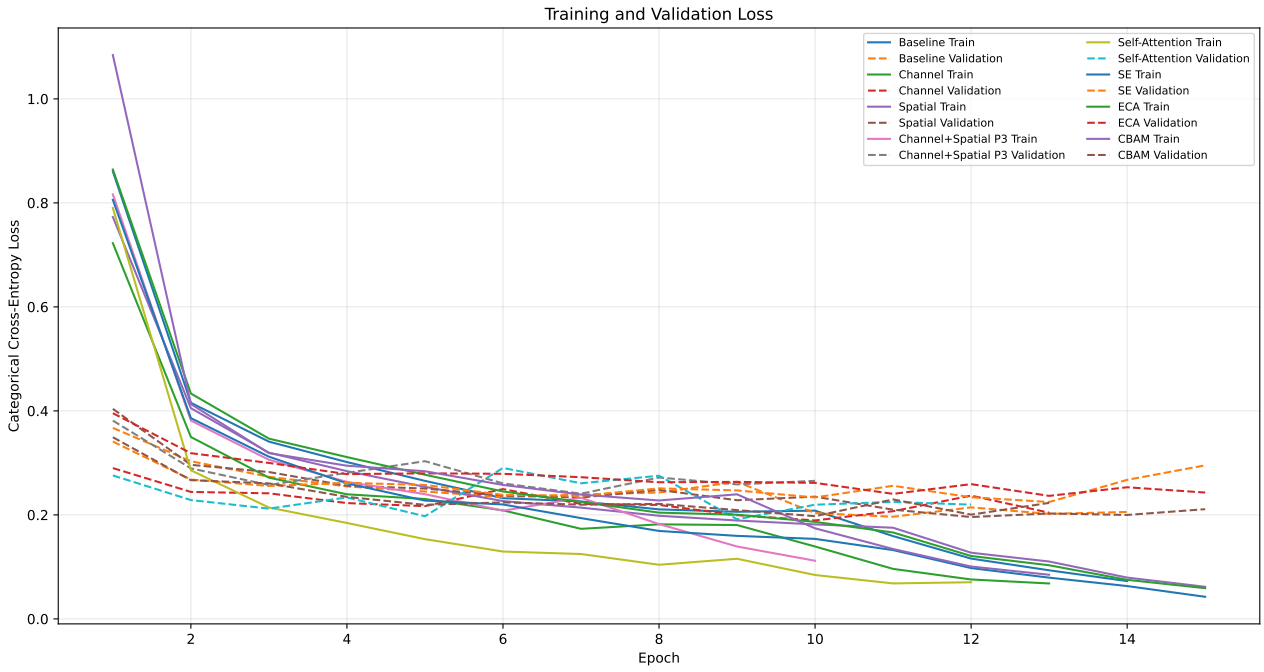

Figure 6: Training and validation categorical cross-entropy loss.

A decreasing training loss indicates that the model is learning the training data. A decreasing validation loss suggests improved performance on unseen validation samples. If the training loss continues to decrease while the validation

loss increases, the model may be overfitting.

The loss curves should be considered together with the accuracy curves because a model can have similar accuracy to another model while exhibiting different confidence and loss behavior.

24.3 Attention Mechanism versus Accuracy

Figure 7 compares the test classification accuracy of the baseline and the attention mechanisms. The values were obtained from the model-comparison results in the notebook.

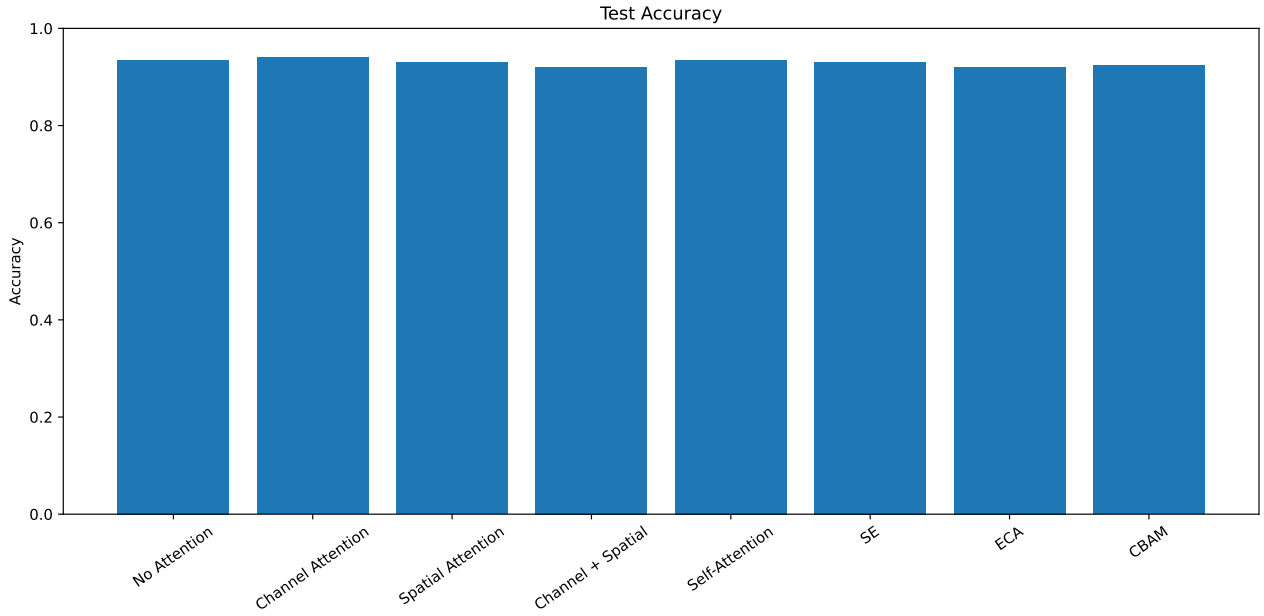

Figure 7: Test accuracy for different attention mechanisms.

The baseline ResNet50 achieved a test accuracy of 0.9347, or 93.47%. Channel attention achieved the highest accuracy among the compared models, with 0.9401, or 94.01%.

Spatial attention achieved 0.9292, while the combined channel and spatial attention model at P3 achieved 0.9201. Self-attention achieved 0.9347, matching the baseline accuracy. These results indicate that the effect of attention depends on the mechanism used; adding attention does not automatically improve classification accuracy.

24.4 Attention Mechanism versus Macro F1-score

Figure 8 compares the macro F1-scores of the evaluated models. Macro F1 calculates the F1-score separately for each class and then takes the unweighted mean across classes.

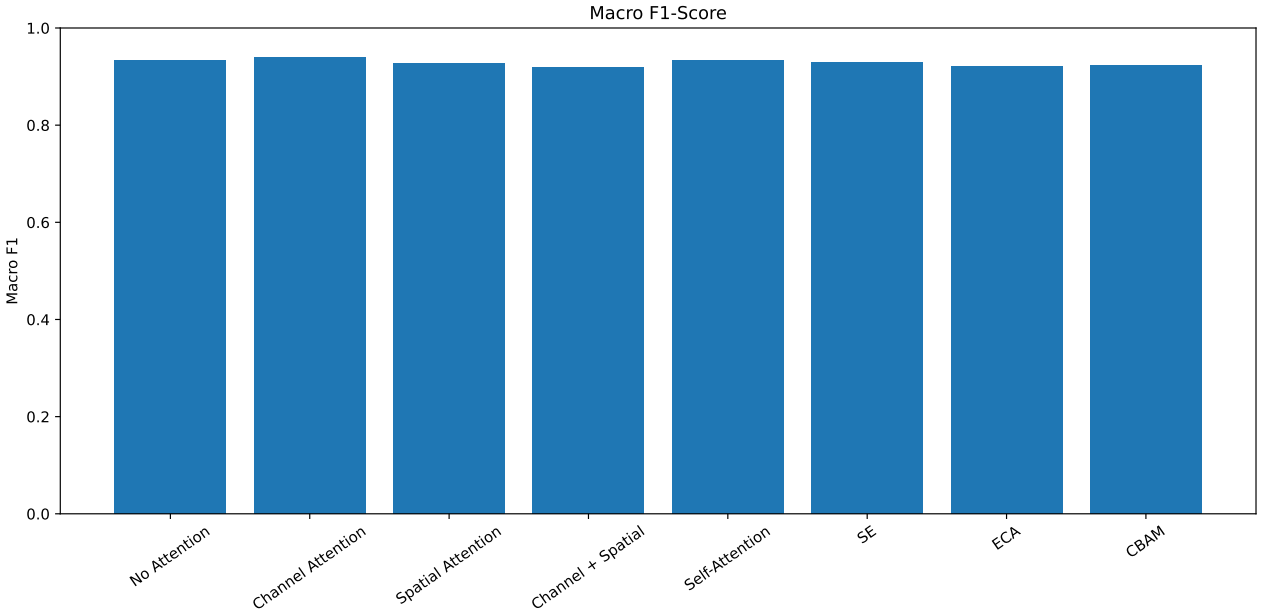

Figure 8: Macro F1-score for different attention mechanisms.

Channel attention achieved the highest macro F1-score of 0.9389, compared with 0.9339 for the baseline. The combined channel and spatial attention model at P3 achieved 0.9194.

The macro F1-score complements accuracy because it gives equal weight to each class, rather than allowing classes with more correct predictions to dominate the overall measure. In this experiment, channel attention performed better than the baseline on both accuracy and macro F1-score.

24.5 Accuracy versus Model Parameters

Figure 9 illustrates the relationship between test accuracy and the total number of model parameters.

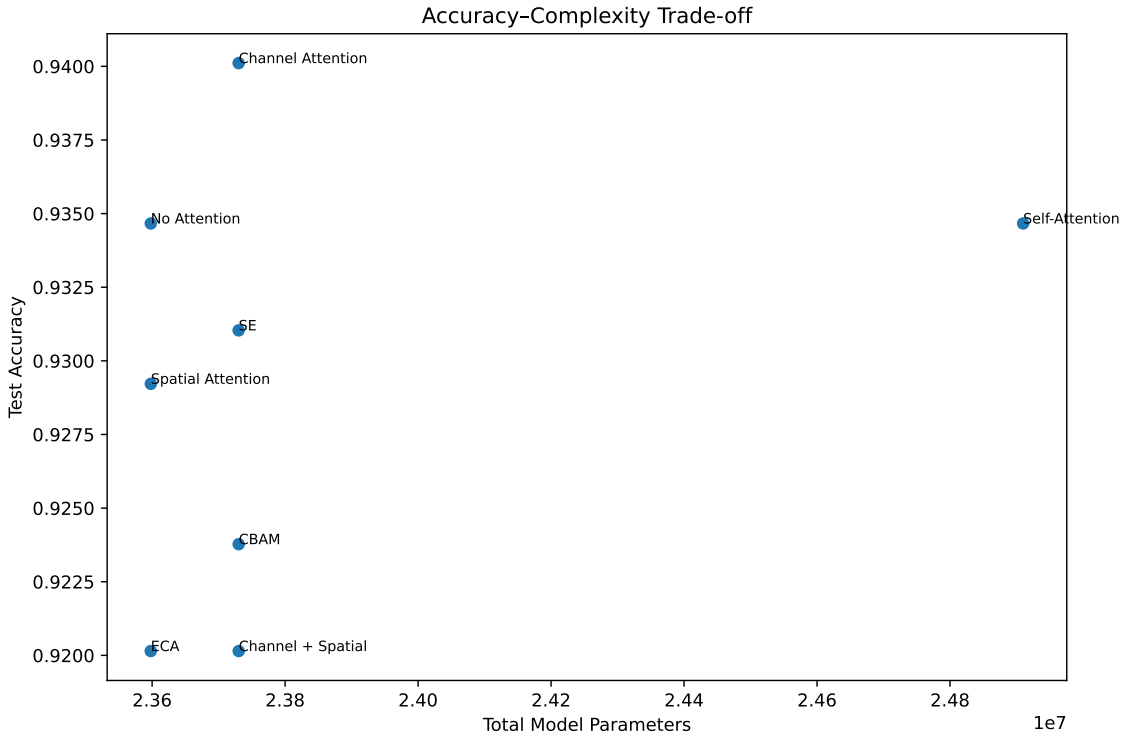

Figure 9: Accuracy–complexity trade-off for the evaluated models.

The baseline contains 23,597,957 parameters, whereas the channel-attention model contains 23,730,117 parameters. Thus, channel attention adds parameters while increasing test accuracy from 93.47% to 94.01%.

Self-attention contains 24,909,958 parameters and achieves the same test accuracy as the baseline. This comparison illustrates why model selection should consider both predictive performance and model complexity rather than accuracy alone.

24.6 Macro F1 versus Inference Time

Figure 10 compares macro F1-score with measured inference time per image.

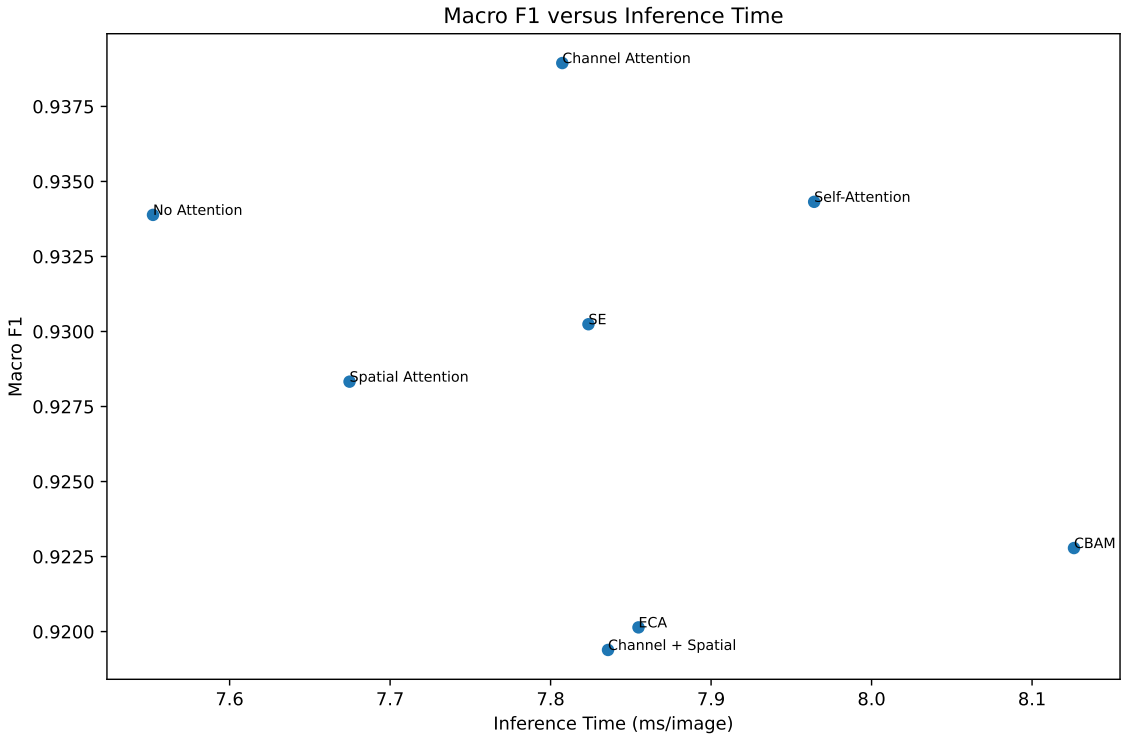

Figure 10: Macro F1-score versus inference time per image.

The baseline inference time was approximately 7.5523 milliseconds per image, while channel attention required approximately 7.8073 milliseconds per image. Channel attention therefore provided a higher macro F1-score, with a small increase in measured inference time.

Self-attention required approximately 7.9642 milliseconds per image, and CBAM required approximately 8.1261 milliseconds per image. These measurements are specific to the execution environment and should not be treated as universal inference times for the architectures.

24.7 Overall Graphical Analysis

The graphical and numerical comparisons show that the attention mechanism has an important effect on the classification results. Channel attention achieved the highest test accuracy and macro F1-score among the models in the Section 19 comparison. However, other mechanisms, including spatial attention and combined channel and spatial attention at P3, did not outperform the baseline in overall test accuracy.

The training and validation curves should be used to discuss convergence and possible overfitting, while the accuracy, macro F1, parameter-count, and inference-time plots support the comparison of predictive performance and computational cost.

25. Confusion Matrix Analysis

Confusion matrices were generated for the baseline ResNet50 and the Channel + Spatial Attention model applied at P3. The matrices compare the true class labels with the predicted class labels for the five flower categories: daisy, dandelion, roses, sunflowers, and tulips.

25.1 Baseline ResNet50

Figure 11 shows the confusion matrix for the baseline ResNet50 model.

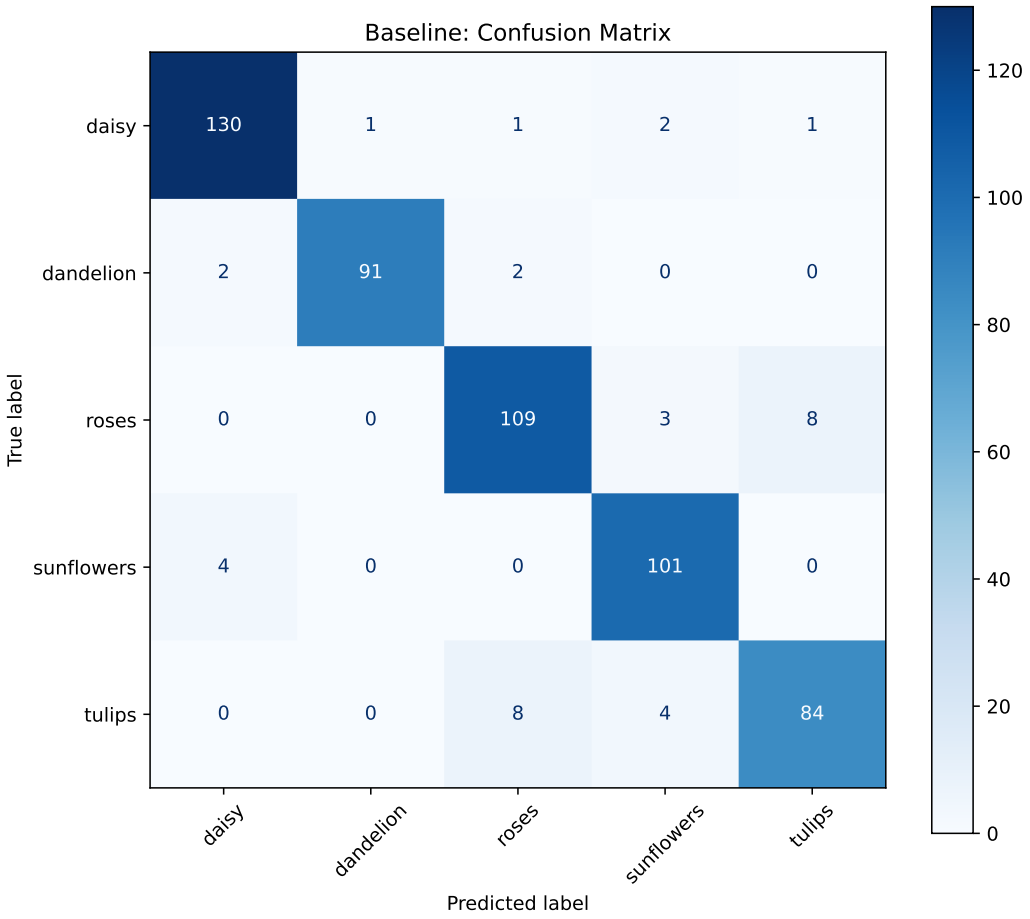

Figure 11: Confusion matrix of the baseline ResNet50 model.

The diagonal entries represent correctly classified images, whereas off-diagonal entries represent misclassifications. The baseline achieved an overall test accuracy of 93.47%.

Based on the per-class classification report, dandelion achieved the highest F1-score of 0.9733, followed by daisy with 0.9594. Tulips had the lowest F1-score among the five classes, at 0.8889, indicating that it was comparatively more challenging for the baseline model.

25.2 Channel + Spatial Attention at P3

Figure 12 presents the confusion matrix for the Channel + Spatial Attention model at P3.

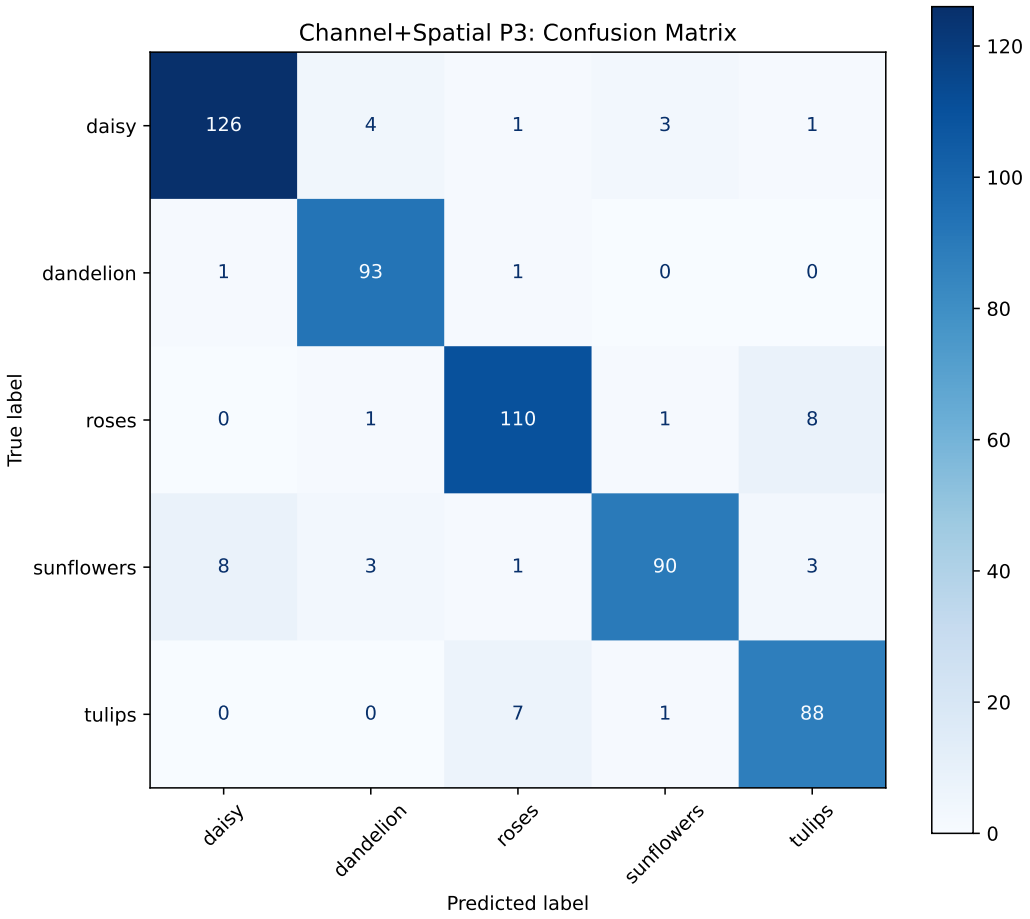

Figure 12: Confusion matrix of Channel + Spatial Attention at P3.

The Channel + Spatial P3 model achieved an overall test accuracy of 92.01%. Its class-wise performance differs from that of the baseline: dandelion recall increased, while sunflower recall decreased. Therefore, the effect of attention was not uniform across all five flower classes.

25.3 Per-Class Performance Comparison

Table 12 compares the per-class precision, recall, and F1-score reported by the notebook. The support column indicates the number of test images belonging to each class.

Table 12: Per-class performance of the baseline and Channel + Spatial Attention at P3.

Model	Class	Precision	Recall	F1-score	Support
Baseline	Daisy	0.9559	0.9630	0.9594	135
Baseline	Dandelion	0.9891	0.9579	0.9733	95
Baseline	Roses	0.9083	0.9083	0.9083	120
Baseline	Sunflowers	0.9182	0.9619	0.9395	105
Baseline	Tulips	0.9032	0.8750	0.8889	96
Channel + Spatial P3	Daisy	0.9333	0.9333	0.9333	135
Channel + Spatial P3	Dandelion	0.9208	0.9789	0.9490	95
Channel + Spatial P3	Roses	0.9167	0.9167	0.9167	120
Channel + Spatial P3	Sunflowers	0.9474	0.8571	0.9000	105
Channel + Spatial P3	Tulips	0.8800	0.9167	0.8980	96

The per-class results show the following observations:

- **Daisy:** The baseline achieved an F1-score of 0.9594, compared with 0.9333 for Channel + Spatial P3.
- **Dandelion:** Recall increased from 0.9579 to 0.9789. However, precision decreased from 0.9891 to 0.9208, and the F1-score decreased from 0.9733 to 0.9490.
- **Roses:** The F1-score increased slightly from 0.9083 to 0.9167.

- **Sunflowers:** Precision increased from 0.9182 to 0.9474, but recall decreased from 0.9619 to 0.8571. The F1-score consequently decreased from 0.9395 to 0.9000.
- **Tulips:** Recall increased from 0.8750 to 0.9167, and the F1-score increased slightly from 0.8889 to 0.8980.

25.4 Interpretation and Conclusion

The confusion matrices and per-class metrics indicate that the attention mechanism affected different flower classes in different ways. Dandelion recall and tulip recall improved with Channel + Spatial P3, while daisy F1-score and sunflower recall decreased. Roses showed a small improvement in F1-score.

Although some individual classes benefited from attention, the overall accuracy decreased from 93.47% for the baseline to 92.01% for Channel + Spatial P3. Therefore, the results do not support the conclusion that this attention configuration improved overall classification performance.

The experiment demonstrates the importance of evaluating attention-based models using both overall metrics and class-wise performance. Confusion matrices help identify misclassification patterns, while precision, recall, and F1-score reveal which classes account for changes in the overall

26. Qualitative Analysis of Attention

Quantitative classification metrics describe how accurately a model predicts the flower classes, but they do not directly explain which image regions contribute to its predictions. Therefore, qualitative visualization was performed using feature maps, attention maps, attention overlays, and Grad-CAM.

The notebook generates visualizations for the baseline ResNet50, channel attention, spatial attention, combined channel and spatial attention, and self-attention models.

26.1 Intermediate Feature Maps

Intermediate feature maps help illustrate how image representations change when attention is applied to convolutional features.

Figure 13 shows the feature-map comparison before and after attention.

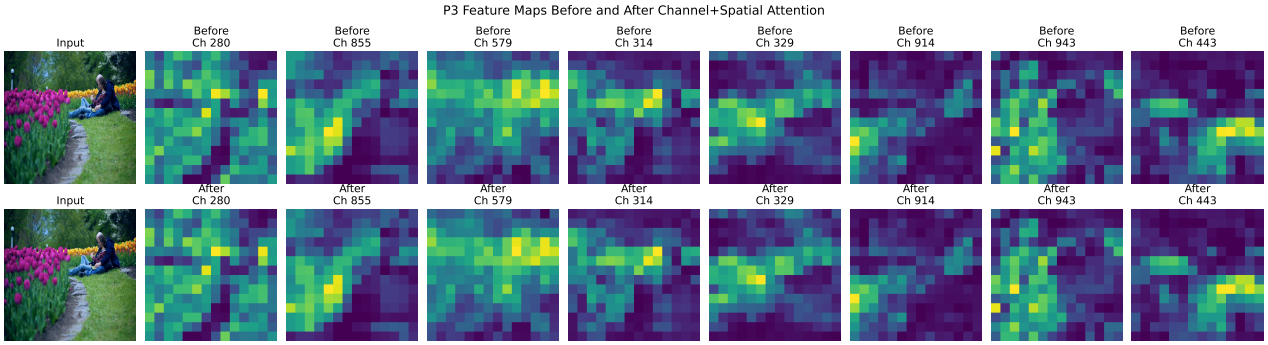

Figure 13: Intermediate feature maps before and after attention.

The feature maps provide a qualitative view of the representations used by the network. Differences in activation patterns indicate that the attention operation modifies the feature representation. However, feature-map intensity alone does not prove that a model uses more meaningful or class-discriminative information.

26.2 Qualitative Comparison Across Models

The same test images were used to compare the baseline and attention-based models. The comparison includes the original image, baseline Grad-CAM, channel-attention Grad-CAM, spatial attention, combined channel and spatial Grad-CAM, and the self-attention response.

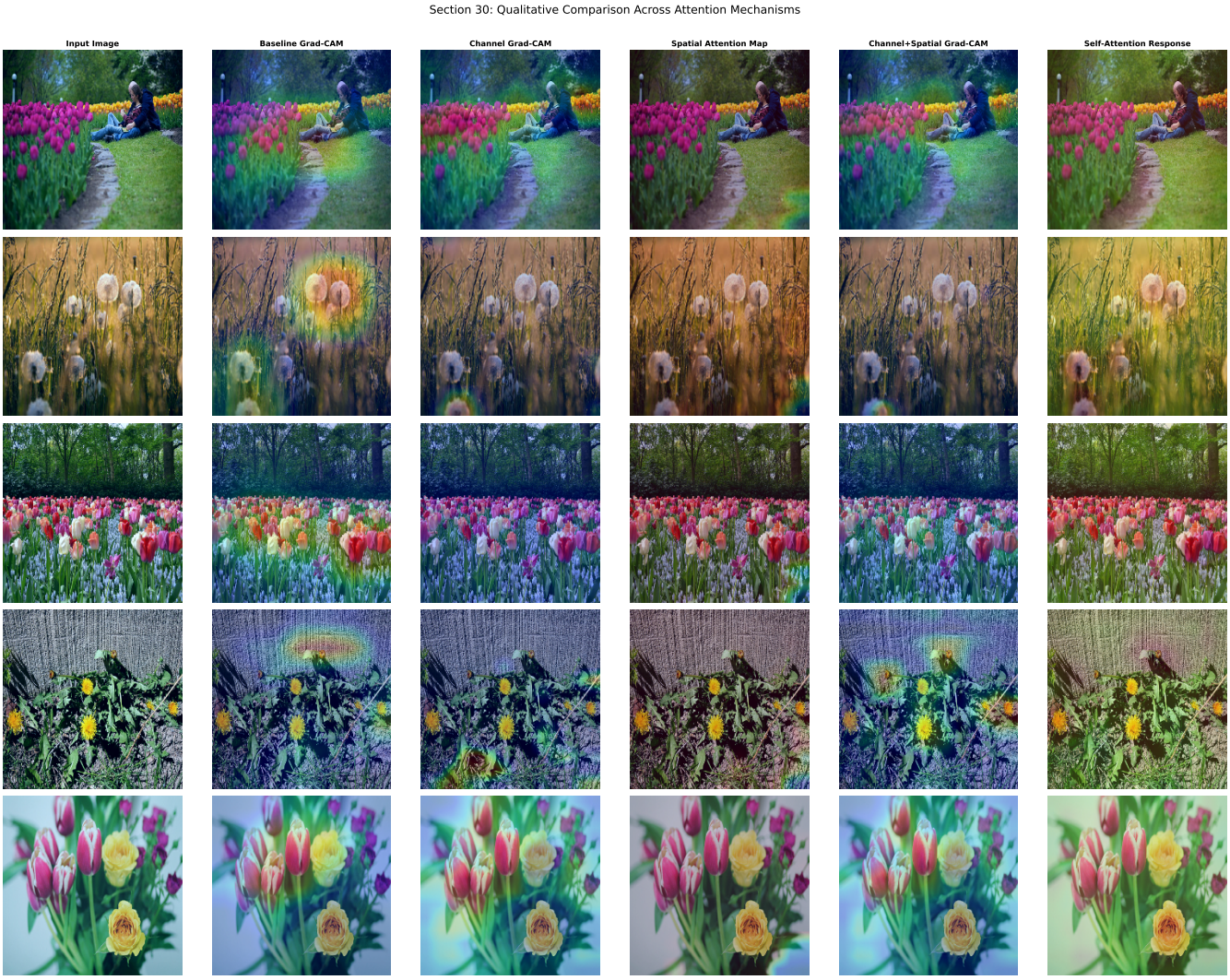

Figure 14: Qualitative comparison of the baseline and attention-based models on selected test images.

The notebook selected three examples correctly classified by the baseline and two examples incorrectly classified by the baseline for this visualization. Comparing the same input across models helps identify differences in their visual responses without confounding the comparison with changes in the input image.

The heatmaps should be interpreted by examining the regions with strong activation and comparing them with the flower and background. An attention map should not automatically be considered better merely because it is brighter or more concentrated.

27. Grad-CAM Visualization

Gradient-weighted Class Activation Mapping (Grad-CAM) is a post-hoc explanation method that highlights image regions associated with a selected class prediction. It uses gradients of a class score with respect to feature maps from a convolutional layer.

Let A^k represent feature map k, and let y^c denote the score for class c. The importance coefficient is calculated as

$$\alpha_k^c = \frac{1}{HW} \sum_{i=1}^{H} \sum_{j=1}^{W} \frac{\partial y^c}{\partial A_{ij}^k}.$$

The Grad-CAM heatmap is

$$L_{\text{Grad-CAM}}^c = \text{ReLU}\left(\sum_k \alpha_k^c A^k \right).$$

The resulting heatmap is normalized, resized to the input image resolution, and overlaid on the original image to facilitate visual interpretation.

27.1 Baseline Grad-CAM

Figure 15 presents the baseline model's Grad-CAM visualization.

Figure 15: Grad-CAM visualization for the baseline ResNet50.

The heatmap indicates the image regions associated with the baseline model's predicted class. The highlighted regions should be compared with the visible flower structure and surrounding background to assess whether the prediction appears to rely on relevant visual evidence.

27.2 Grad-CAM Comparison Across Models

The notebook generated Grad-CAM visualizations for the baseline, channel attention, spatial attention, and combined channel and spatial attention models.

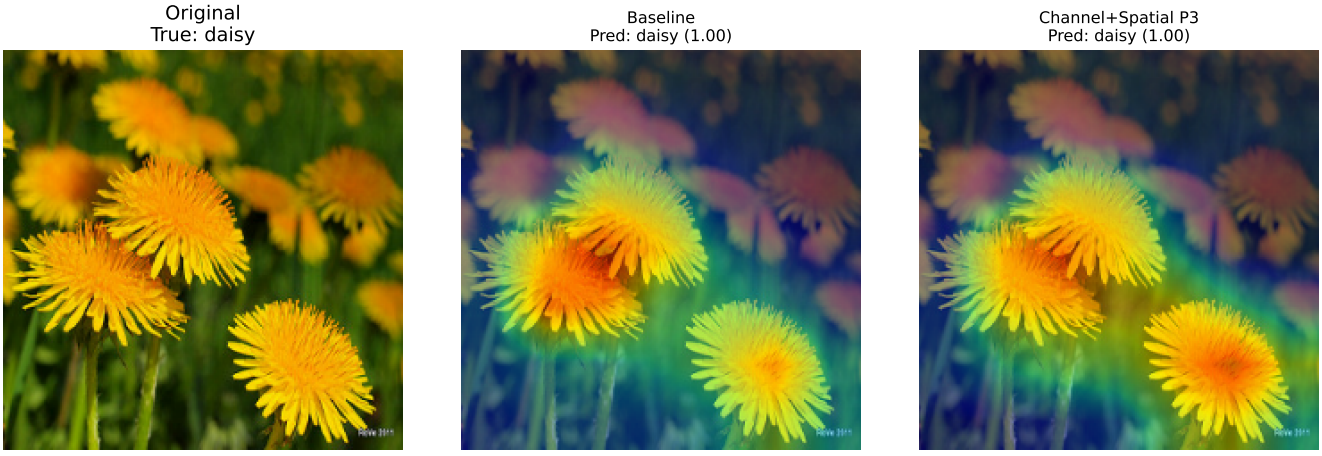

Figure 16: Grad-CAM comparison across the baseline and attention-based models.

For the sample image used in the Grad-CAM implementation, all four models predicted the class *roses*. The maximum predicted probabilities reported by the notebook were:

Table 13: Predictions for the sample used in Grad-CAM analysis.

Model	Maximum Predicted Probability
Baseline ResNet50	0.9957
Channel Attention	0.9995
Spatial Attention	0.9995
Channel + Spatial Attention	0.9987

These values describe the model outputs for this particular sample. They do not establish that the attention-based models are more accurate overall. A high predicted probability also does not guarantee that the highlighted image regions provide a faithful or complete explanation.

27.3 Grad-CAM Interpretation

The heatmaps should be examined for the following visual patterns:

- Activation over petals and the central flower structure.
- Activation over leaves and other background regions.
- Differences in the spatial distribution of activation between the baseline and attention-based models.
- Changes in the highlighted regions when the predicted class is incorrect.

The notebook also generated visualizations for examples where the baseline and attention model disagreed.

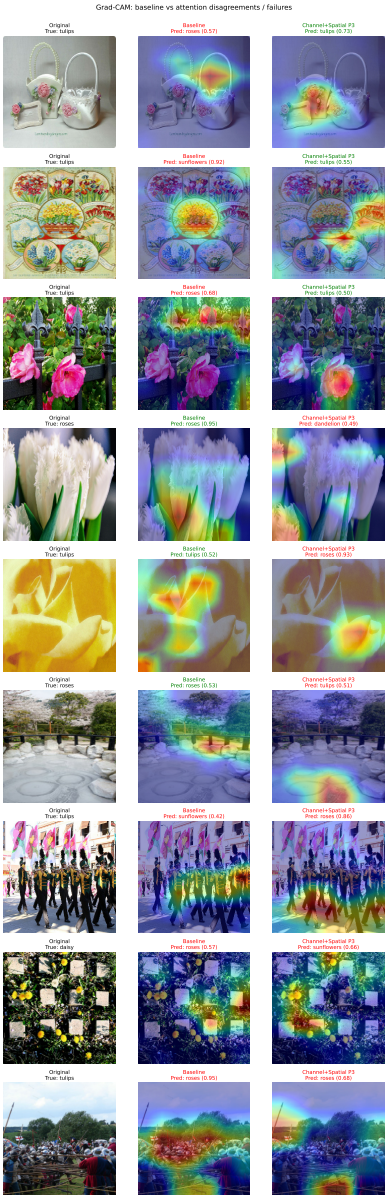

Figure 17: Grad-CAM analysis of examples where model predictions differed.

The notebook reported five cases where both models were correct, three cases where attention corrected a baseline error, three cases where attention produced an error for a case the baseline handled correctly, and three cases where both models were incorrect.

These counts describe the analyzed subset, not the entire test set. They show that the effect of attention can vary from one image to another. Therefore, Grad-CAM comparisons should be interpreted together with classification metrics and the actual predictions.

28. Internal Attention Maps

Internal attention and post-hoc explanation are different concepts. An attention module computes weights during the model's forward pass and uses them to modify feature representations. Grad-CAM uses gradients after the model has produced a prediction.

The following subsections examine the internal channel, spatial, and self-attention visualizations generated by the notebook.

28.1 Channel-Attention Visualization

Channel attention assigns importance weights to feature channels. For a feature tensor with C channels, the attention weights can be represented as

$$M_c(F) = [m_1, m_2, \ldots, m_C].$$

The notebook extracts the learned channel weights and displays the top 20 channels in descending order of their weights.

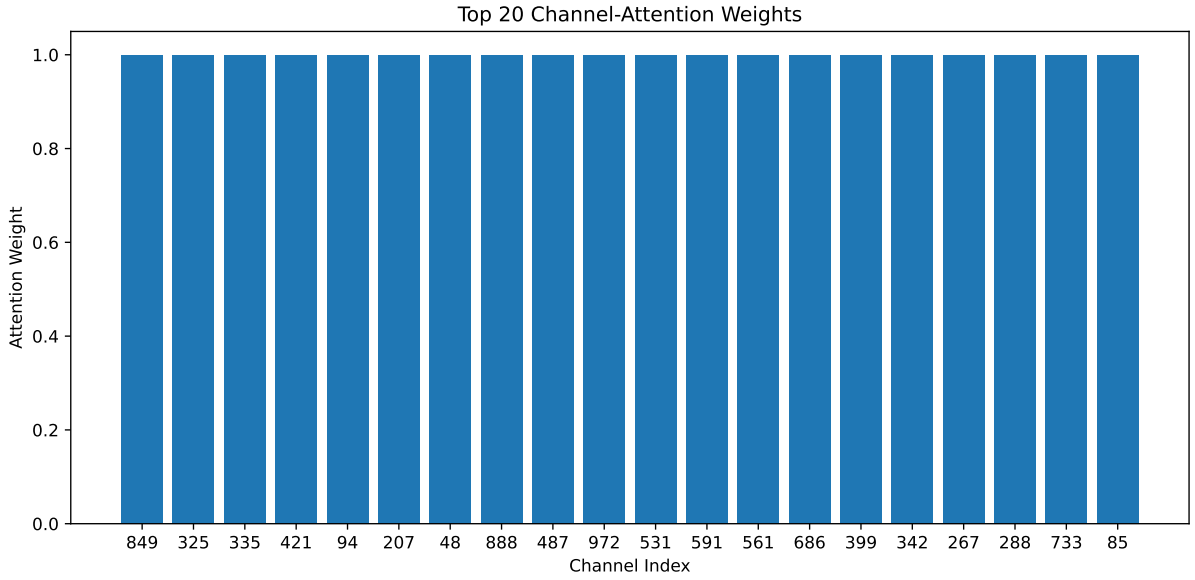

Figure 18: Top 20 channel-attention weights for the selected test image.

The plot allows the relative importance weights of the selected channels to be compared. A channel with a higher weight receives greater multiplicative emphasis from this attention operation. However, a high weight does not by itself establish that the channel represents a particular flower feature.

This visualization is based on the selected input image. Different images may produce different channel weights.

28.2 Spatial-Attention Visualization

Spatial attention assigns weights to spatial locations in a feature map. The attention map can be represented as

$$M_s(F) \in \mathbb{R}^{H \times W}.$$

The notebook extracts the spatial-attention maps and resizes them to 224×224 for comparison with the original image.

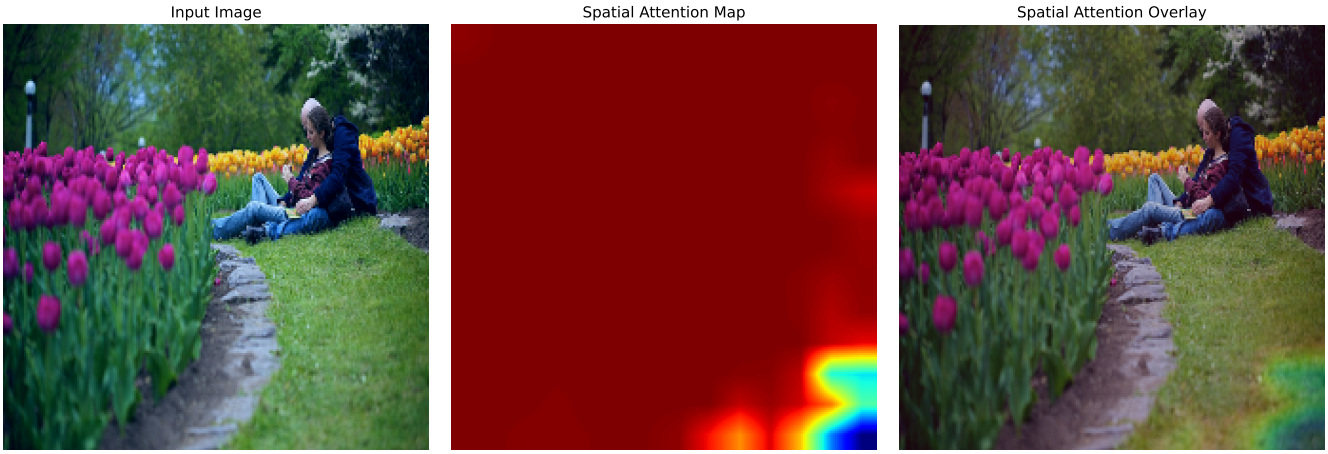

Figure 19: Spatial-attention maps and overlays for the selected flower image.

The spatial map shows how attention weights vary across image locations. The overlay helps relate these weights to the visible flower and background.

Regions with higher displayed attention values receive greater weight in the spatial-attention operation. The interpretation should be based on the actual map: the visualization alone does not establish whether the model has completely suppressed the background or focused exclusively on petals.

28.3 Self-Attention Visualization

Self-attention enables interactions between feature locations. To visualize its effect, the notebook computes the mean absolute difference between the feature tensors before and after the self-attention layer:

$$R(i,j) = \frac{1}{C}\sum_{c=1}^{C} \left| F_{\text{after}}(i,j,c) - F_{\text{before}}(i,j,c) \right| .$$

The resulting response is resized to the input resolution and displayed alongside the input image and an overlay.

Figure 20: Aggregated self-attention response and its overlay on the input image.

This figure visualizes where the feature representation changed most in magnitude after the self-attention operation. It is an aggregated response map, not the full query-key attention matrix. Consequently, it should not be interpreted as a direct display of every interaction between distant spatial locations.

29. Attention Position Visualization

ResNet50 produces feature maps at different spatial resolutions and semantic levels. The notebook compares spatial-attention visualizations at P2, P3, and P4 using the same test image.

The feature-map resolutions before resizing are:

- P_2: $28 \times 28 \times 512$
- P_3: $14 \times 14 \times 1024$
- P_4: $7 \times 7 \times 2048$

The spatial-attention maps are resized to 224×224 so that their displayed locations can be compared with the input image.

Figure 21: Spatial-attention visualization at ResNet50 stages P2, P3, and P4 for the same test image.

The visualization can be used to study how the spatial response changes across feature stages. Earlier stages preserve finer spatial detail, while deeper stages have lower spatial resolution and encode more abstract features. Resizing a low-resolution attention map does not recover information that was lost at the original feature-map resolution.

The following observations should be made from the generated figure:

- Whether the P2 map contains more spatially detailed responses than the deeper-stage maps.
- Whether the P3 map highlights identifiable flower structures.
- Whether the P4 map produces a broader or more localized response.
- Whether the high-response regions overlap the flower or extend into the background.

These are criteria for interpreting the plotted results, not assumptions that every map must exhibit the same behavior.

The notebook's position-ablation results also require careful interpretation: differences in classification accuracy across P2, P3, and P4 do not necessarily arise only from attention position if the corresponding training or fine-tuning procedures differ.

29.1 Summary of Qualitative Analysis

The visualizations provide complementary information about the models. Grad-CAM highlights regions associated with a selected class score, channel attention displays feature-channel weights, spatial attention displays location-dependent weights, and the self-attention response summarizes changes in feature activations.

No single visualization is sufficient to establish that an attention mechanism is universally superior. The qualitative observations should therefore be interpreted together with test accuracy, macro F1-score, class-wise metrics, and the confusion matrices presented in the preceding sections.

30. Qualitative Comparison Across Attention Mechanisms

Qualitative comparison was performed to investigate how different attention mechanisms influence the image regions and feature representations used by ResNet50. The analysis complements the quantitative evaluation by examining individual predictions and their corresponding visual explanations.

The notebook includes examples of correctly classified images and examples where the baseline and attention-based models disagree.

30.1 Comparison of Correctly Classified Images

Figure 22 presents examples used to compare the visual responses of the models. The same input image is compared across the available visualization methods so that differences in the highlighted regions can be examined.

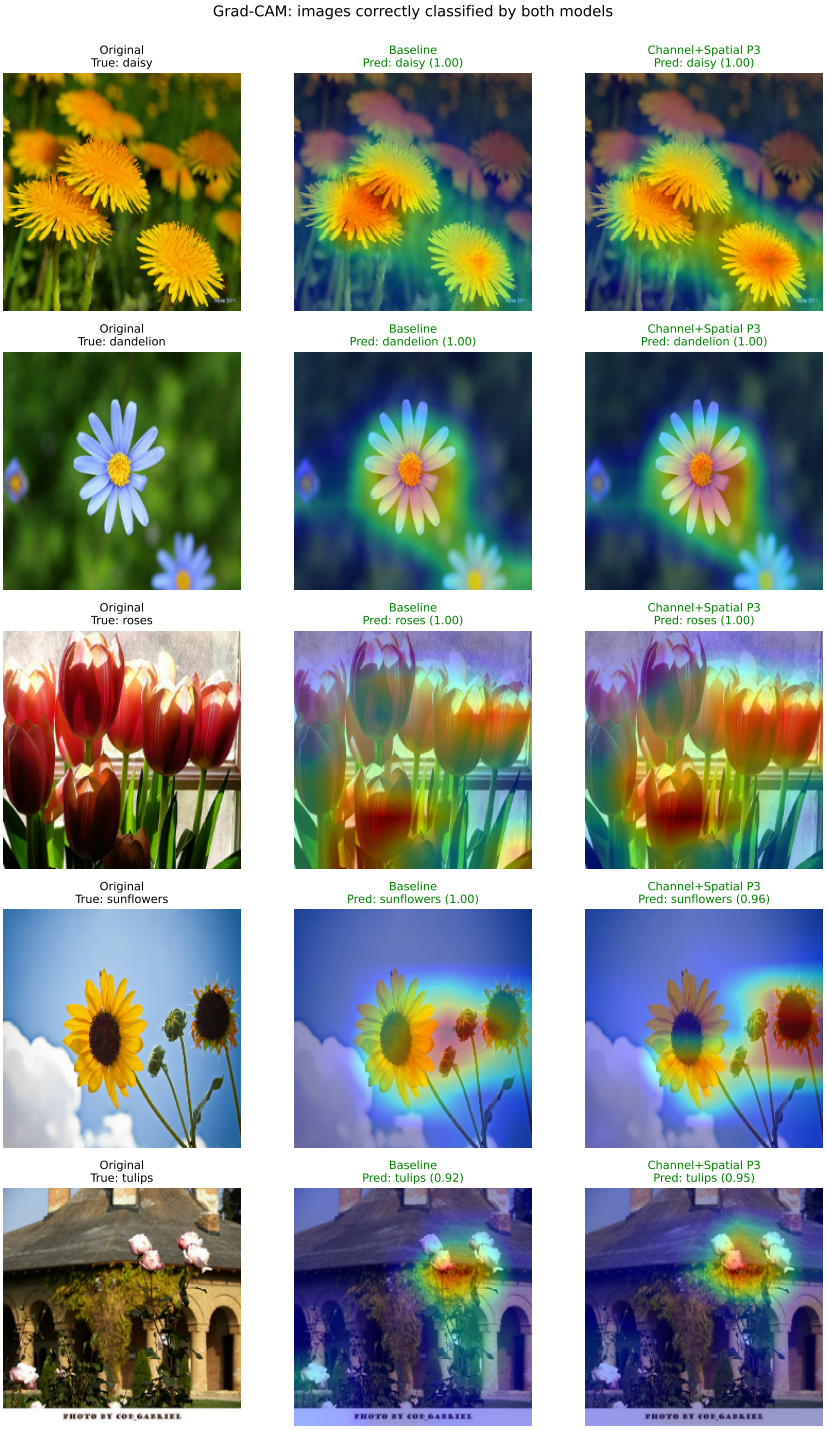

Figure 22: Grad-CAM visualizations for selected correctly classified examples.

For each example, the highlighted regions should be compared with the visible flower structure. Strong activation over the flower may indicate that the model uses relevant image information, whereas strong activation over unrelated background objects requires further investigation.

A heatmap should not be judged only by its brightness or concentration. The prediction, the original image, and the spatial distribution of the activation must be considered together.

30.2 Comparison of Incorrect and Disagreeing Predictions

Figure 23 shows the notebook's Grad-CAM comparisons for examples where the models produced different outcomes.

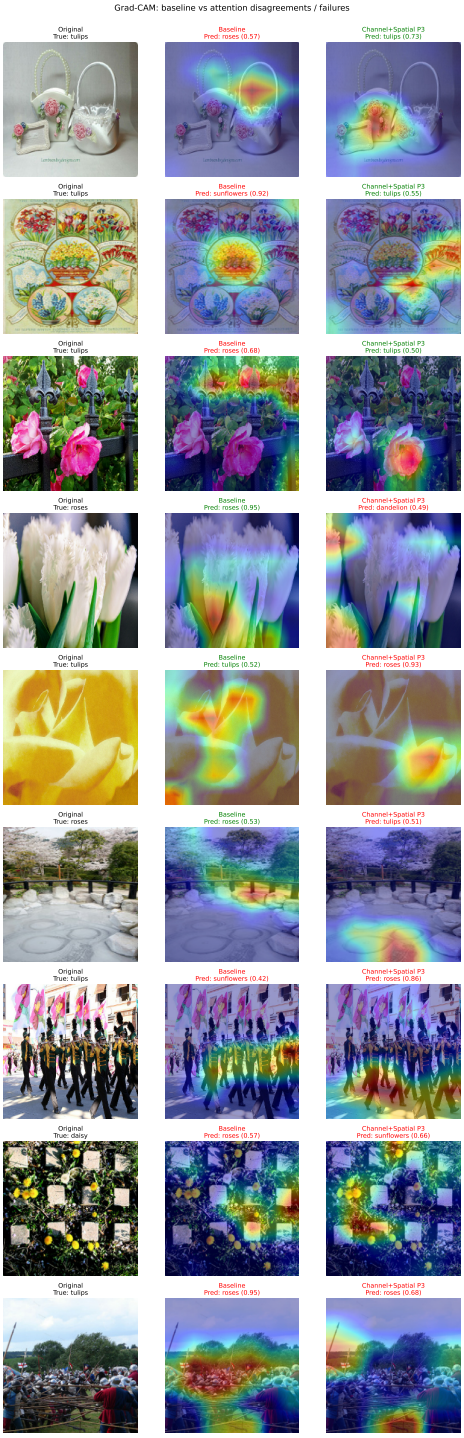

Figure 23: Grad-CAM analysis of examples where model predictions differed.

The notebook reported the following counts for the analyzed subset:

- Both models correct: 5 cases.
- Attention corrected a baseline error: 3 cases.
- Attention produced an error where the baseline was correct: 3 cases.
- Both models incorrect: 3 cases.

These counts describe the selected examples and should not be interpreted as test-set-wide performance measurements. They demonstrate that attention can help on some images while reducing performance on others.

30.3 Qualitative Criteria

The visual comparisons were examined using the following criteria:

- **Foreground localization:** whether activation overlaps the flower rather than unrelated image regions.
- **Background response:** whether leaves, soil, other flowers, or background objects receive strong activation.
- **Petal and center localization:** whether the heatmap covers petals or the central flower structure.

- **Activation coverage:** whether the response covers a small region or a larger part of the flower.
- **Incorrectly attended regions:** whether the response emphasizes irrelevant structures.
- **Prediction consistency:** whether visual differences correspond to changes in the predicted class.
- **Across-image consistency:** whether similar patterns appear for different input images.

The visualizations support qualitative observations, but they do not prove that attention is causally responsible for a prediction. Such claims require additional controlled tests.

31. Required Visualizations

The following table summarizes the visual evidence included in the report and the corresponding figure files.

Table 14: Visualization checklist and corresponding figure files.

Visualization	Figure file
Sample flower images	`sample_flower_images.pdf`
Training and validation accuracy	`section24_training_validation_accuracy.pdf`
Training and validation loss	`section24_training_validation_loss.pdf`
Baseline confusion matrix	`section25_baseline_confusion_matrix.pdf`
Attention-model confusion matrix	`section25_channel_spatial_p3_confusion_matrix.pdf`
Feature maps before and after attention	`section26_feature_maps_before_after_attention.pdf`
Channel-attention weights	`section28_channel_attention_top20.pdf`
Spatial-attention map and overlay	`section28_spatial_attention_visualization.pdf`
Self-attention response	`section28_self_attention_response.pdf`
Baseline Grad-CAM	`section27_baseline_gradcam.pdf`
Grad-CAM across models	`section27_gradcam_all_models.pdf`
Grad-CAM for disagreeing predictions	`section27_gradcam_disagreements.pdf`
Attention at P2, P3, and P4	`section29_attention_position_visualization.pdf`
Accuracy comparison	`section19_accuracy.pdf`

The listed files are available in the supplied `Figures/` directory. The checklist identifies the visualization evidence for the experiment; it does not imply that every possible attention mechanism has been implemented or evaluated.

32. Recommended Visualization Layout

Figure 24 provides a combined visualization of the input and the model responses. It helps compare the original image with the spatial patterns generated by the available attention and explanation methods.

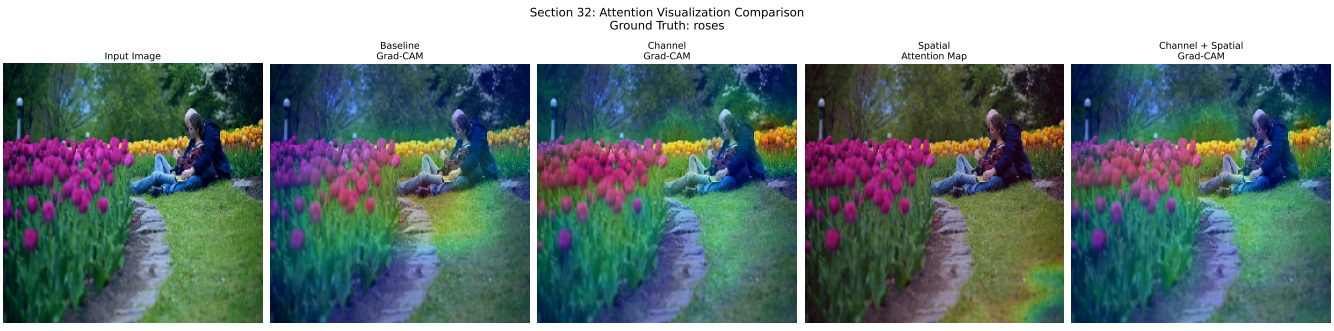

Figure 24: Combined visualization of the input image and attention-related responses.

The preferred comparison layout is:

Input	Baseline Grad-CAM	Channel Grad-CAM	Spatial Attention Map	Channel+Spatial Grad-CAM

When comparing SE, ECA, CBAM, and self-attention, the same input image should be used wherever the corresponding model outputs are available. The explanation method must be stated clearly: a spatial-attention map is

not the same as a Grad-CAM heatmap, and an aggregated self-attention response is not necessarily the original attention matrix.

33. Required Experimental Analysis

The experiment should be assessed using three complementary forms of analysis.

33.1 Quantitative Analysis

The quantitative evaluation includes:

$$\boxed{\text{Accuracy} + \text{Precision} + \text{Recall} + \text{Macro F1}}$$

Accuracy measures the fraction of correctly classified test images. Precision measures the reliability of positive predictions for a class, while recall measures the fraction of actual examples of that class that were correctly detected. Macro F1 averages the class-wise F1-scores without weighting classes by their sample counts.

33.2 Complexity Analysis

The computational evaluation includes:

$$\boxed{\text{Total Parameters} + \text{Inference Time}}$$

Parameter count provides a measure of model size, while inference time estimates prediction latency under the tested environment. Inference time depends on hardware, batch size, framework, warm-up, and measurement procedure. Therefore, comparisons should use the same evaluation environment and timing method.

33.3 Qualitative Analysis

The qualitative evaluation includes:

$$\boxed{\text{Grad-CAM} + \text{Attention Maps} + \text{Visual Interpretation}}$$

Grad-CAM helps identify image regions associated with a selected class score. Internal attention maps show the weights computed by attention modules, while feature-map visualizations show changes in intermediate representations.

The overall evaluation is therefore:

$$\boxed{\text{Quantitative} + \text{Computational} + \text{Qualitative}}$$

No single measure is sufficient to establish that one model is universally better than another.

34. Experimental Protocol

Fair comparisons require the experimental conditions to remain consistent wherever possible. The notebook uses a fixed, stratified train-validation-test split with random seed 42, containing 2,569 training images, 550 validation images, and 551 test images, for a total of 3,670 images across five classes.

The image input size is 224×224, and the models use ResNet50 with ImageNet preprocessing. These settings provide a common foundation for the experiments.

For each controlled ablation, the following conditions should be kept identical wherever the experiment permits:

- training, validation, and test images;
- random seed and data split;
- input resolution and preprocessing;
- data augmentation;
- backbone architecture;
- classifier head;
- optimizer and learning-rate schedule;
- batch size and number of epochs;
- fine-tuning strategy; and
- evaluation metrics.

Only the factor being investigated should change. For example, an attention-position experiment should change the position of the attention module while keeping the other settings consistent.

If training schedules or fine-tuning procedures differ between experiments, those differences should be reported because they can affect the results independently of attention.

35. Minimum Mandatory Ablation Set

The laboratory's minimum experiment set includes the following comparisons:

1. Baseline pretrained ResNet50.
2. Channel attention at P_3.
3. Spatial attention at P_3.
4. Channel + spatial attention at P_3.
5. Channel + spatial attention at P_2.
6. Channel + spatial attention at P_4.
7. Self-attention at P_3.
8. Grad-CAM comparison between the baseline and a selected attention model.
9. Spatial-attention visualization at P_2, P_3, and P_4.
10. Graphical comparison of model performance.

The position-ablation and multi-position plots provide additional visual evidence for the effect of attention placement.

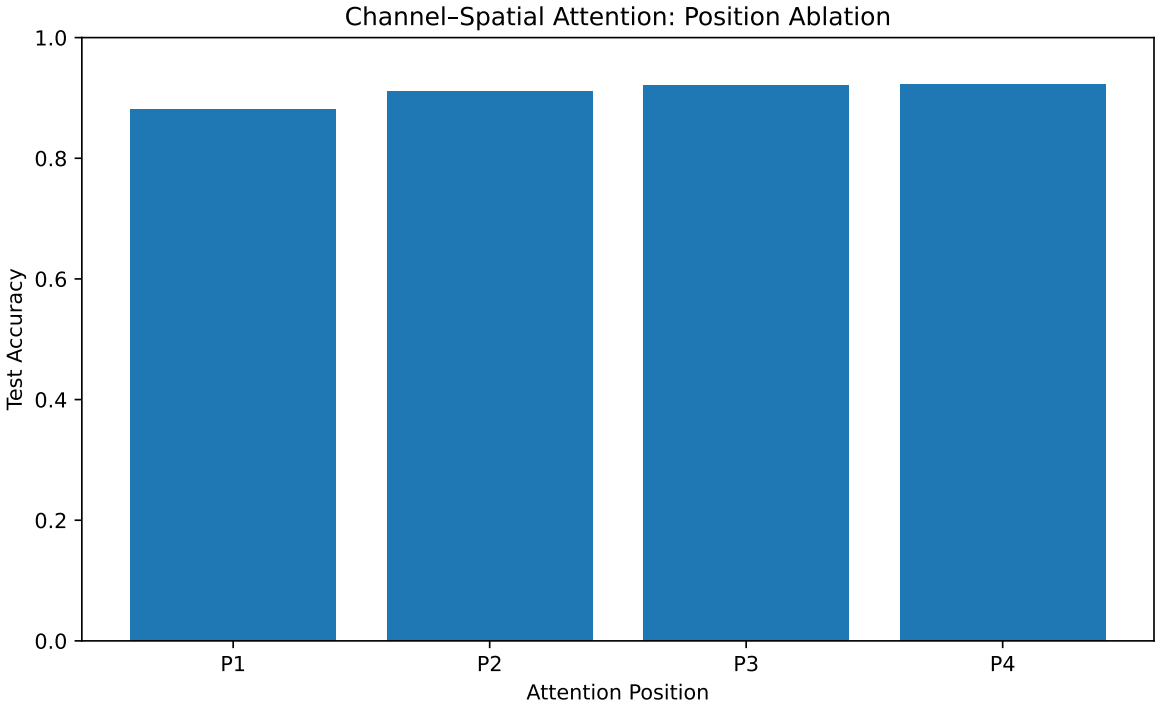

Figure 25: Classification accuracy for the attention-position ablation.

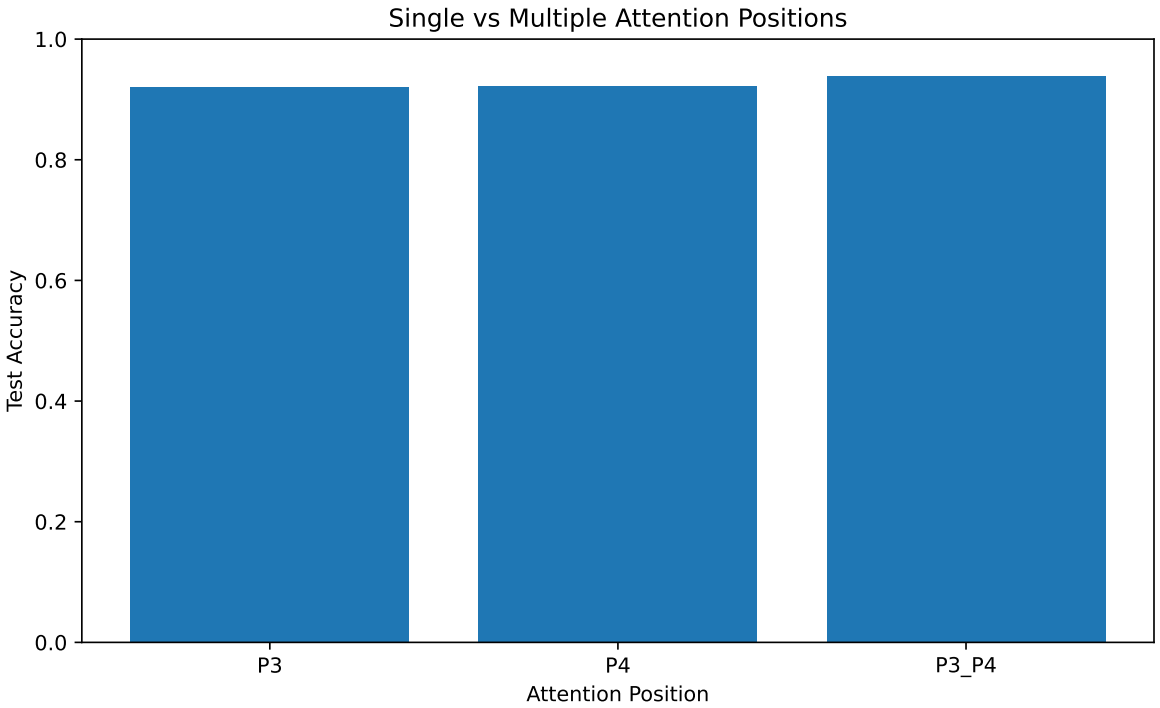

Figure 26: Classification accuracy for the multi-position attention comparison.

These plots should be interpreted using the experimental conditions associated with each result. In particular, a position comparison is not a fully controlled ablation if training or fine-tuning procedures also change.

The minimum set is intended to investigate two main factors:

Attention Type and Attention Position.

The report should distinguish experiments completed in the notebook from experiments that are listed as requirements but have not yet been run.

36. Additional Student Exploration

Additional experiments may investigate mechanisms such as:

SE, ECA, CBAM, Self-Attention

Coordinate Attention may also be explored if an implementation is added. It should not be reported as an experimental result unless it has actually been implemented, trained, and evaluated.

The notebook includes a comparison of several attention mechanisms. Figure 27 presents the corresponding optional comparison plot.

Figure 27: Comparison of the additional attention mechanisms evaluated in the notebook.

For a meaningful comparison, the dataset split, preprocessing, backbone, classifier, training configuration, and attention position should be kept consistent wherever possible.

The additional investigation should report both quantitative and qualitative evidence:

Quantitative Comparison + Qualitative Comparison.

Quantitative results establish differences in predictive performance and computational cost. Qualitative visualizations help examine whether the models respond to similar or different regions of the input. Together, these results provide a more complete evaluation of attention-based image classification.

37. Source Code

The complete implementation for this experiment on image classification using a pretrained ResNet50 model and different attention mechanisms is available in the GitHub repository provided below. The implementation includes the baseline model, channel attention, spatial attention, combined channel and spatial attention, self-attention, and additional attention experiments.

The repository also contains the training and evaluation procedures, performance comparisons, attention visualizations, Grad-CAM analysis, and other experimental results associated with the study.

GitHub Repository Link

https://github.com/AI-MANIAC007/Hands-On-Deep-Learning/

38. Questions Students Must Answer

1. **What is the purpose of residual connections in ResNet50?**
   Residual connections allow the input of a block to be added directly to its output. Instead of learning the complete mapping from scratch, the block learns a residual function. This helps gradients flow through the network during backpropagation and makes it easier to train a deep architecture such as ResNet50.
2. **What does conv2_x represent compared with conv5_x?**
   The conv2_x stage is an earlier stage of ResNet50, whereas conv5_x is the final convolutional stage. In the standard architecture, conv2_x produces relatively high-resolution feature maps that retain fine spatial details. By conv5_x, the feature maps have a lower spatial resolution but contain more abstract information useful for recognizing objects and their semantic characteristics.
3. **Why does the spatial resolution decrease with network depth?**
   The network uses downsampling operations, including strided convolutions and pooling in the early stages. These operations reduce the height and width of the feature maps. Although some spatial detail is lost, the reduction decreases computational cost and allows deeper layers to process a larger effective region of the input image.
4. **Why does the number of feature channels increase with network depth?**
   As the spatial resolution decreases, the network can use more channels to represent different patterns and combinations of features. Early layers may respond to edges or simple textures, while deeper layers combine these patterns into more complex representations. Increasing the channel count provides the capacity needed for these richer representations.
5. **What is meant by low-level, intermediate, and high-level features?**
   Low-level features include edges, colors, corners, and simple textures. Intermediate features can represent combinations of these patterns, such as petal boundaries and repeated flower structures. High-level features represent more complex semantic information that helps distinguish one flower category from another. These descriptions are general tendencies; the features are learned jointly rather than being assigned fixed meanings.
6. **What is the purpose of an attention mechanism?**
   An attention mechanism assigns different weights to features or spatial locations so that their contributions are not treated equally. In image classification, this can help the network emphasize information that is useful for distinguishing classes. However, attention does not guarantee better predictions, because the learned weights depend on the data, model, and training process.
7. **What does channel attention attempt to learn?**
   Channel attention learns a weight for each feature channel. A channel may respond to a particular pattern or combination of visual features, and the attention module adjusts how strongly that channel contributes to the next computation. The weights are learned during training or fine-tuning, depending on the experimental setup.
8. **What does spatial attention attempt to learn?**
   Spatial attention learns weights associated with different locations in a feature map. These weights determine which spatial positions receive greater emphasis. For flower classification, the module may give higher weights to locations containing useful flower structures, but this behavior must be checked from the resulting attention maps.
9. **How is channel attention different from spatial attention?**
   Channel attention operates along the channel dimension and adjusts the importance of feature maps. Spatial attention operates across the height and width dimensions and adjusts the importance of locations. Channel

attention asks which feature channels are useful, whereas spatial attention asks where useful information is located.

10. **Why can channel and spatial attention be used sequentially?**
    The two mechanisms focus on different dimensions of a feature tensor. Channel attention can first recalibrate the feature channels, after which spatial attention can assign weights to the locations in the recalibrated representation. This gives the network an opportunity to refine both the type of feature information and its spatial distribution. The extra operation is useful only if it improves the overall result.
11. **Why might channel attention behave differently at conv3_x and conv5_x?**
    The two stages produce features at different depths. Features at conv3_x generally retain more spatial detail and represent intermediate patterns, while conv5_x contains more abstract representations at a lower spatial resolution. Consequently, the channel weights can emphasize different information at the two stages. The exact difference depends on what the model learns from the training data.
12. **Why can spatial attention be useful at an intermediate ResNet stage?**
    An intermediate stage retains more spatial detail than the final convolutional stage while already representing meaningful structures. Spatial attention at this stage can therefore adjust the importance of locations without relying only on the coarse spatial information available in deeper layers. Whether this improves classification must be determined experimentally.
13. **Why is self-attention computationally expensive when the feature map has high spatial resolution?**
    Self-attention compares spatial positions with one another. If a feature map has height H and width W, the number of spatial positions is $N = HW$. A full attention operation constructs an $N \times N$ matrix, so its memory and pairwise computation grow quadratically with the number of positions. This can make self-attention expensive at high spatial resolutions.
14. **What happens to the size of the self-attention matrix when the spatial resolution increases?**
    The matrix has $N^2 = (HW)^2$ entries. For example, a 14×14 feature map has 196 spatial positions and therefore requires an attention matrix with $196^2 = 38{,}416$ entries per attention head. A 28×28 feature map has 784 positions, giving $784^2 = 614{,}656$ entries per head. Doubling both spatial dimensions therefore increases the number of matrix entries by a factor of 16.
15. **How does attention performance vary when the insertion position is changed?**
    In this experiment, the reported accuracies for the channel-plus-spatial attention position comparison were 0.9111 at P_2, 0.9201 at P_3, and 0.9220 at P_4. Thus, P_4 achieved the highest accuracy among these three reported results, followed closely by P_3. The differences should be interpreted cautiously because the fine-tuning procedures were not fully identical across all position experiments.
16. **How does attention at P_2 differ visually from attention at P_4?**
    The P_2 representation has a higher spatial resolution and can preserve finer details, such as edges, petal boundaries, and texture. The P_4 representation has a lower spatial resolution and contains more abstract features. As a result, the corresponding attention maps may differ in their level of detail and coverage. The actual visual difference should be judged from the generated P_2, P_3, and P_4 maps rather than assumed in advance.
17. **Does applying attention at multiple stages always improve classification performance?**
    No. Additional attention modules increase the number of operations and may change the feature representations in ways that are not helpful for the task. The notebook's multi-position experiment reported an accuracy of 0.9383 for the $P_3 + P_4$ configuration, compared with 0.9201 for P_3 and 0.9220 for P_4 in the position comparison. This suggests that using multiple positions can be beneficial in that experiment, but it does not establish that adding more attention modules will always improve accuracy.
18. **Does Channel→Spatial attention behave differently from Spatial→Channel attention?**
    Yes. In Channel→Spatial attention, the channel weights are applied before spatial attention processes the resulting features. In Spatial→Channel attention, the order is reversed. Since each operation receives a different intermediate representation, the final outputs can differ. In the reported comparison, Spatial→Channel achieved an accuracy of 0.9274 and macro F1-score of 0.9258, while Channel→Spatial achieved an accuracy of 0.9201 and macro F1-score of 0.9194.
19. **How does the spatial-attention kernel size influence the attention map?**
    The kernel size controls the local spatial neighborhood used to calculate the attention response. A smaller kernel focuses on nearby patterns, while a larger kernel can incorporate information from a broader neighborhood. In the reported comparison, the 3×3 kernel achieved an accuracy of 0.9274 and macro F1-score of 0.9263. The 5×5 and 7×7 settings both achieved an accuracy of 0.9310 and macro F1-score of 0.9304. The larger kernels did not improve these reported metrics over each other, and their inference times should also be considered.
20. **How does SE attention differ from ECA attention?**
    Squeeze-and-Excitation (SE) attention summarizes each channel using global spatial information and uses a small bottleneck network to calculate channel weights. Efficient Channel Attention (ECA) avoids the dimensionality-reduction bottleneck and uses a lightweight one-dimensional convolution to model local interactions between neighboring channel descriptors. ECA is designed to reduce the additional parameter cost while retaining channel recalibration.
21. **How does CBAM combine channel and spatial attention?**

Convolutional Block Attention Module (CBAM) applies channel attention and spatial attention sequentially. Channel attention first recalibrates the feature channels. Spatial attention then uses information aggregated across channels to calculate a two-dimensional spatial weighting map. The two operations allow the module to adjust both channel importance and spatial location importance.

22. **How does self-attention differ from channel attention?**
Channel attention assigns weights to feature channels, whereas self-attention models relationships between feature positions. For a spatial feature map, self-attention can allow one location to incorporate information from other locations, including distant ones. Channel attention does not directly calculate a full pairwise relationship between every spatial position.
23. **Does the attention mechanism with the largest number of parameters necessarily provide the highest accuracy?**
No. A larger model has more parameters, but additional parameters do not guarantee that the model will learn more useful features. In the reported model comparison, channel attention achieved an accuracy of 0.9401, while self-attention achieved 0.9347 despite having more total parameters. This shows why parameter count and classification performance must be evaluated separately.
24. **What is the difference between an internal attention map and Grad-CAM?**
An internal attention map is generated by the attention module during the forward pass and is used to reweight features. Grad-CAM is a post-hoc visualization calculated using gradients of a selected class score with respect to convolutional feature maps. An internal attention map shows the weights used by the module, whereas Grad-CAM highlights regions associated with a particular class prediction.
25. **Why is Grad-CAM called a post-hoc explanation technique?**
Grad-CAM is called post-hoc because it is applied after a trained model has been produced; it does not have to be part of the model's original prediction mechanism. Gradients are calculated for a selected class, and the resulting weighted feature maps are combined to create a heatmap. The visualization helps inspect the prediction but does not prove the model's complete reasoning.
26. **Does the attention model suppress irrelevant background regions?**
This cannot be concluded for every image from the accuracy score alone. The Grad-CAM and spatial-attention visualizations must be examined to determine whether the response over the background is weaker than in the baseline. In some images the attention model may emphasize useful flower regions, while in others it may still respond strongly to leaves or background objects.
27. **Are the attention maps of correctly classified and incorrectly classified images different?**
They can be different because the input image, predicted class, and learned feature responses affect the visualization. Correctly classified images may show activation over useful flower structures, whereas an incorrect prediction may involve activation over confusing background patterns or non-discriminative regions. However, this is not a universal rule; the actual maps should be compared for each selected example.
28. **Which flower classes benefit most from attention?**
The overall accuracy and macro F1-score do not identify the class that benefits most. To answer this question, the baseline and attention-model confusion matrices and class-wise recall should be compared. The class showing the largest improvement in recall, supported by the corresponding confusion-matrix counts, would be a suitable candidate. A specific class cannot be identified reliably without those class-wise results.
29. **What information can be obtained from a confusion matrix that cannot be obtained from overall accuracy alone?**
A confusion matrix shows the number of images from each actual class that were predicted as each class. It reveals which flower categories are confused with one another and whether errors are concentrated in particular classes. Overall accuracy reduces all predictions to a single number and therefore hides the types of mistakes the model makes.
30. **Why is macro F1 useful in addition to accuracy?**
Accuracy measures the overall proportion of correct predictions, while macro F1 calculates an F1-score for each class and then takes their unweighted mean. This gives every class equal importance, regardless of how many examples it contains. Macro F1 is useful when class-wise performance differs because a strong result on common classes cannot completely conceal poor performance on another class.
31. **What is meant by the accuracy–complexity trade-off?**
The accuracy–complexity trade-off describes the balance between classification performance and the resources needed by a model. A model may achieve higher accuracy but require more parameters, memory, or computation. A smaller model with slightly lower accuracy may be preferable when computational resources are limited. The appropriate choice depends on the application's accuracy and efficiency requirements.
32. **What is meant by the F1–inference-time trade-off?**
This trade-off compares macro F1-score with the time required to process an image. A model may improve class-wise performance but take longer to make predictions because of its additional operations. Plotting macro F1 against inference time helps determine whether the performance gain is worth the extra latency for the intended application.
33. **Why must every ablation study use the same dataset split?**

Different train-test splits can produce different results because some images may be easier or harder to classify than others. If two models are evaluated on different test images, the difference in performance may be caused by the data rather than the attention mechanism. Keeping the same split makes the comparison more controlled and meaningful.

34. **Why must the same training protocol be maintained when comparing different attention mechanisms?**
    Training settings influence how well a model learns. Differences in the optimizer, learning rate, augmentation, number of epochs, or fine-tuning strategy can change the final performance even when the architecture is unchanged. Keeping these settings consistent helps isolate the effect of the attention mechanism. If a setting must differ, the difference should be documented as a limitation of the comparison.
35. **What conclusions can be drawn from the Grad-CAM comparison?**
    The Grad-CAM comparison shows how the baseline and attention-based models distribute their class-associated activation across the same input images. The notebook's selected examples include cases where attention corrected a baseline error and cases where it introduced an error. Therefore, the effect of attention was not uniform across the analyzed images. Grad-CAM provides useful visual evidence, but its conclusions should be considered alongside the test metrics and confusion matrices.

39. Expected Outcome

At the end of the experiment, students should understand that an attention mechanism is not an isolated module whose behavior is independent of the backbone.

Its effectiveness depends on:

$$\boxed{\begin{aligned}\text{Attention Effectiveness} = {} & \text{Attention Type} \\ & + \text{Attention Position} \\ & + \text{Feature Resolution} \\ & + \text{Feature Semantics} \\ & + \text{Model Complexity.}\end{aligned}}$$

Students should also understand that classification performance alone does not provide a complete explanation of an attention mechanism.

A comprehensive analysis should therefore combine:

$$\boxed{\text{Classification Performance}}$$

with

$$\boxed{\text{Computational Complexity}}$$

and

$$\boxed{\text{Qualitative Visualization.}}$$

Therefore:

$$\boxed{\text{Complete Attention Analysis} = \text{Quantitative} + \text{Computational} + \text{Qualitative}}$$

The final conclusions must be based on the student's experimental results rather than on assumptions about which attention mechanism should perform best.

40. References

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